

Estimating properties of semi-resolved globular clusters in M31: A forward modelling and deep learning framework

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ABSTRACT

Recent and upcoming wide-field photometric surveys will observe hundreds of thousands of globular clusters (GCs) in the next decade. This poses the emerging challenge of deriving reliable estimates of their properties using methods that scale to large volumes of data. We present a new framework that combines forward modelling with deep learning, allowing accurate and detailed analysis of GCs. To this end, we develop CREMANT, a forward-modelling code that is used to create realistic mock images of GCs from state-of-the-art N -body simulations, and embed them in real backgrounds. We generate mock images resembling real HST observations of M31 (PHAT and PHAST surveys) and use them to train a neural network, π -DOC, to decontaminate GC images from field stars, reconstruct 2D dynamical mass maps, and infer global properties (age, distance, ellipticity, and position angle) from multi-band photometric images and pixel colour-magnitude diagrams.

On synthetic data, π -DOC constrains cluster mass and age to within $^{+0.30}_{-0.43}$ dex and $^{+2.73}_{-3.57}$ Gyr at 2σ — a conservative confidence range that remains comparable to typical M31 literature uncertainties and only $\sim 1.5\times$ wider than typical mass uncertainties for fully resolved Galactic GCs. Decontamination remains robust even for GCs that are several magnitudes fainter than the backgrounds. Distance ($^{+79}_{-79}$ kpc) and ellipticity ($^{+0.07}_{-0.08}$) estimates are comparatively more uncertain, with the latter primarily exacerbated by faint and only mildly elliptical GCs.

Applied to 349 M31 GCs in the PHAT and PHAST footprints, the model shows agreement with the existing literature within a factor of four for total masses, while efficiently scaling to large volumes of data with an inference speed of ~ 10 GCs per second. We release a publicly available catalogue of π -DOC's estimates for M31, all our training data, as well as both codes for CREMANT and π -DOC, for direct use to the community. This work illustrates that the combination of forward modelling and deep learning is a scalable and physically motivated approach for inferring GC properties from large multi-band photometric datasets.

Key words. methods: numerical, deep learning – globular clusters: general – galaxies: star clusters: general – objects: M31

1. Introduction

Globular clusters (GCs) are dense, gravitationally bound stellar systems composed of up to a few million stars. They are among the oldest known stellar populations in the Universe, with Galactic GCs showing ages of up to 13 Gyr (Leaman et al. 2013; Valcin et al. 2025, 2026). Thanks to recent gravitationally lensed observations with the James Webb Space Telescope (JWST), the progenitors of present-day GCs have been identified at redshifts corresponding to only about 500 Myr after the Big Bang (Vanzella et al. 2023; Adamo et al. 2024; Mowla et al. 2024; Claeysens et al. 2026). While their exact origin still remains an open question, their formation is linked to key phases of structure formation in the early Universe (e.g. high pressure star formation and galactic merger events Kruijssen 2015; Lahén et al. 2020, 2025; Taylor et al. 2025). Consequently, the properties of nearby GCs provide unique fossil records of the physical conditions under which galaxies assembled during the first billion years of cosmic history (Forbes & Bridges 2010; Forbes et al. 2018; Forbes 2020).

Over their roughly 13 Gyr of evolution, GCs are shaped by internal dynamical processes, such as two-body relaxation and stellar evolution, as well as by external effects, such as the coupling to the tidal field of their host galaxy (Spitzer 1987; Ves-

perini & Heggie 1997; Baumgardt & Makino 2003; Gieles et al. 2011; Trenti & van der Marel 2013; Watkins et al. 2015; Bianchini et al. 2016, 2026; Kuzma et al. 2016; Webb & Vesperini 2016; Malhan et al. 2018; Jindal et al. 2019; Marklund et al. 2026). Related to the possible formation channel of GCs during a galaxy's assembly phase, some GCs are also thought to have been accreted during galaxy merger events (Forbes & Bridges 2010; Renaud et al. 2017; Forbes et al. 2018; Forbes 2020; Massari et al. 2019; Myeong et al. 2019; Pfeffer et al. 2021; Chen & Gnedin 2024), or even be the remnant nuclei of dwarf galaxies (e.g. Ibata et al. 1995; Alfaro-Cuello et al. 2019; Clontz et al. 2024; Pagnini et al. 2025, 2026). The interplay between these mechanisms and the resulting complex evolution over a Hubble-time, make GCs useful testbeds for many different astrophysical topics: stellar evolution models, dynamical theories of collisional systems, and the long-term effects of galactic environments on stellar populations.

The development of high-precision N -body (Aarseth 2003) and Monte Carlo codes (such as the MOCCA and CMC codes Hypki & Giersz 2013; Giersz et al. 2013; Rodriguez et al. 2016), and particularly GPU-accelerated implementations such as Nbody6++GPU (Wang et al. 2015), has made it possible to simulate clusters containing up to and above a million stars, reaching

parity with the mass regime of typical Galactic GCs (e.g. Wang et al. 2016; Arca sedda et al. 2024; Bianchini et al. 2026; Marklund et al. 2026). These modern simulations also include ingredients such as time-dependent tidal fields (Renaud et al. 2011; Webb et al. 2024) and stellar evolution (Kamleh et al. 2022; Bissekenov et al. 2025; Wu et al. 2026), offering increasingly realistic representations of GCs’ complex dynamics, such as the presence of dark remnants or of internal rotation. Despite these advances, such direct simulations remain extremely demanding in terms of computational time and energy consumption, limiting their use for large-scale parameter studies or statistical comparisons (see e.g. Bianchini et al. 2026). These computational challenges make deep learning an attractive complementary approach, capable of leveraging computationally expensive simulation suites whose sampling of parameter space is necessarily sparse.

In parallel, the growth of deep-field and wide-field surveys are pushing us into a new era for GC observations. Space- and ground-based facilities such as the Hubble space telescope (HST), JWST (Gardner et al. 2023), the Euclid mission (Euclid Collaboration et al. 2022), and the Vera C. Rubin Observatory (LSST, Ivezić et al. 2019), are rapidly increasing both the number of known clusters and the redshift range over which they can be studied. Recent projections estimate that Euclid alone could detect several hundred thousand GCs across the local Universe (Euclid Collaboration et al. 2025). Upcoming missions, including the Roman space telescope (Mosby et al. 2020), are expected to continue this trend by delivering deep, high-resolution multi-band imaging over large areas of the sky. These developments underline an emerging challenge: the need for scalable, automated tools capable of analysing vast samples of clusters efficiently.

Deriving the fundamental physical properties of GCs, including their masses, ages, metallicities, and structural parameters, is essential for using them as probes of star cluster evolution, galaxy assembly, and the formation history of the local Universe. Yet obtaining accurate estimates of these properties for large samples remains challenging, particularly at extragalactic distances. Dynamical masses derived from kinematic data (Bellini et al. 2017) become increasingly difficult to obtain beyond the Local Group, and a fixed mass-to-light (M/L) ratio — often assumed in its place — is not strictly valid for GCs (as demonstrated in e.g. Baumgardt 2017; Bianchini et al. 2017). Constructing dedicated grids of N -body simulations (Baumgardt 2017) avoids this assumption but remains computationally expensive; Monte-Carlo based methods such as MOCCA (Hypki & Giersz 2013; Giersz et al. 2013) are cheaper and better suited to large grid explorations, at the cost of more stringent assumptions (e.g. spherical symmetry). Age and metallicity are typically estimated via isochrone fitting of colour-magnitude diagrams (CMDs; Rosenberg et al. 1999; Zoccali et al. 2003; Dotter et al. 2010; Leaman et al. 2013; Valcin et al. 2025, 2026), a comparatively robust approach that is nonetheless not easily scalable to the data volumes expected from next-generation surveys.

In this context, machine learning, and in particular deep learning (DL), offers promising alternatives (Bialopetravičius & Narbutis 2020a,b; Chardin & Bianchini 2021; Hiegel et al. 2023; Guiglion et al. 2024). DL approaches have already shown great potential in automating the classification of extragalactic star clusters (Zhang et al. 2025), estimating their ages (Boin et al. 2026; Viaña et al. 2026), dynamical properties (Pasquato & Chung 2016; Askar et al. 2019; Bialopetravičius et al. 2019; Pasquato et al. 2024), and estimating global parameters of GCs directly from imaging data (Chardin & Bianchini 2021). The

latter introduced the proof-of-concept algorithm π -DOC (Predicting Images for the Dynamics Of globular Clusters), trained on mock photometric observations generated from forward-modeled N -body simulations (Miholics et al. 2016; Bianchini et al. 2017). The study demonstrated that deep networks can infer underlying physical quantities of GCs, such as their internal mass distribution, age, and distance, directly from photometric images. This approach offers not only an efficient and scalable method, but is also one of the first tools capable of mapping a two-dimensional mass distribution without assuming a mass-to-light (M/L) ratio.

The initial proof-of-concept algorithm relied on idealised mock images of ground based observations constructed from low-mass GC simulations ($< 50k$ images), lacking realistic observational uncertainties. Neither instrumental noise nor galactic field-star contaminants were modeled. Moreover, the limited parameter coverage restricted applicability to just five Galactic GCs. Such limitations reflect a synthetic gap: a mismatch between idealised training data and complex real-world observations that typically degrades performance. Nevertheless, these results demonstrated the strong potential of DL approaches for large-scale analyses of GCs.

In this work, we address the limitations of the proof-of-concept version of π -DOC. We extend the underlying data with state-of-the-art simulations of realistic GCs from both the ROLLIN’ suite (Bianchini et al. 2026; Marklund et al. 2026) and MOCCA simulations (Askar et al. 2025; Zhao et al. 2026). We also move beyond ground based observations of Galactic GCs, and consider partially resolved GCs in the Andromeda galaxy (M31) using archival HST data. To generate realistic training data, we develop CREMANT, a forward-modelling framework that transforms numerical simulations into realistic mock observations by combining synthetic cluster images with real observations of field stars (referred to as backgrounds), simultaneously modelling instrumental and observational effects. These steps substantially reduce the synthetic gap between the training data and real observations. The underlying data and the construction of realistic mock images are described in Section 2, and the code for CREMANT is publicly available online¹.

Building upon the original π -DOC architecture, we develop a unified multi-task network that is applied on multi-band photometric images and that learns to decontaminate those input images (e.g. removal of the background), map the corresponding 2D dynamical mass distribution, and infer the cluster’s age, distance, ellipticity, and position angle. A full description of the inputs, outputs, and the network’s new architecture, can be found in Section 3. The code for π -DOC is also publicly available online².

The remainder of the paper is structured as follows. In Section 4 we assess the performance of π -DOC using realistic mock observations, exploring biases and its robustness to noise. In Section 5 we assess the reliability of π -DOC’s estimates as a function of their representational similarity to the training data. In Section 6, we apply our algorithm to 349 GCs in M31, present the main results, and compare π -DOC’s estimates with the literature. We conclude with a discussion regarding possible extensions and improvements for π -DOC in Section 7, and our conclusions are found in Section 8.

¹ <https://github.com/arnmarklund/cremant>

² <https://github.com/arnmarklund/pidoc>

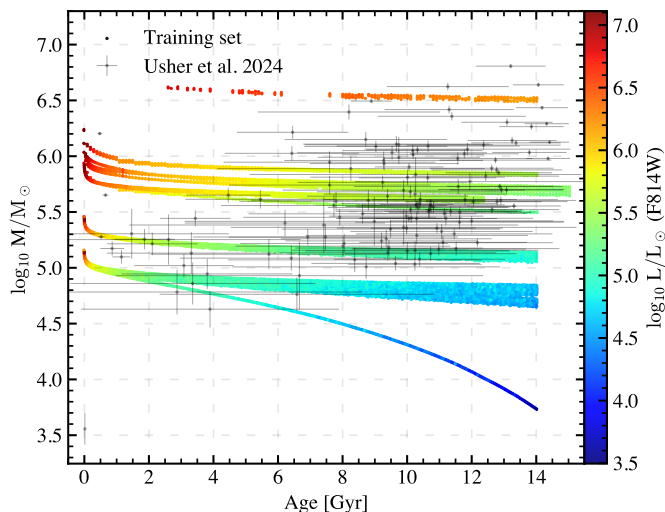


Fig. 1. Comparison of the mass–age relation for our mock images and the M31 GCs from (Usher et al. 2024) located within the PHAT and PHAST footprints. The mock images are colour-coded by their total luminosity in the F814W passband, while the observed GCs are shown as black points. Our mock sample broadly covers the M31 GC systems in the mass versus age parameter space. Nonetheless, some gaps remain, particularly at the highest masses.

2. Real and synthetic data

π -DOC aims to automate the detailed analysis of GCs directly from multi-band photometric observations. As a supervised neural network, its training data must be both physically relevant and representative of real observations. We first describe the real observational data used (Section 2.1), then explain how these data inform the construction of realistic synthetic mock images (Sections 2.2 and 2.3), reducing the synthetic gap between simulations and observations. We summarise the data sets used throughout the paper in Section 2.4.

2.1. Real GC data

In this work we consider GCs in M31, using archived HST data from the Panchromatic Hubble Andromeda Treasury (PHAT) survey³ (Dalcanton et al. 2012; Williams et al. 2014, 2023), as well as the recent extension to the southern parts of M31 (PHAST, Chen et al. 2025). The PHAT and PHAST surveys each cover about a third of M31’s star forming disk, divided into 23 and 13 bricks, respectively. The former is observed in six passbands: F110W, F160W, F275W, F336W, F475W and F814W, whereas the PHAST survey only has the latter two available for mosaic images. We consider all bricks, but owing to the low signal-to-noise we choose to only work with the F336W (where applicable), F475W, and F814W passbands (Williams et al. 2023). We use drizzled images (multiple exposures), and align them north to east, conserving the flux and assuming a common scale of 0.04 arcsec/pixel using the `reproject` package⁴. We select GCs from the Peacock et al. (2010a,b) catalogue that lie within the boundaries of the PHAT and PHAST footprints, amounting to a total of 349 GCs. Every image of a GC we consider has a $12'' \times 12''$ field of view (FoV), which is large enough to include typical GCs in M31 to a few times the half-light radii (typical

half-light radii of M31 GCs are $\sim 1\text{--}2''$ Barmby & Huchra 2001; Huxor et al. 2014) at a distance of ~ 785 kpc (McConnachie et al. 2005).

In addition, we sample every brick and keep a total of 12264 images (also with size $12'' \times 12''$) from the PHAT survey that are free of GCs and other star clusters, with the intention of using these as backgrounds⁵ to our synthetic dataset (see Section 2.3 for details).

2.2. Simulated data and the forward modelling algorithm CREMANT

To construct our data set, we must ensure that the synthetic data share both the underlying physical properties and the observational characteristics of real images. We address the first concern by considering two different suites from state-of-the-art simulations:

- (i) The first and main one, refers to the ROLLIN’ suite (Bianchini et al. 2026; Marklund et al. 2026), and consists of initially rotating, axisymmetric direct N -body simulations with an initial amount of 250k–1.5M stars with a metallicity of $[\text{Fe}/\text{H}] = -1.3$ dex, including prescriptions for stellar evolution and an external tidal field. These initially rotating models lead to a wide range of GC-morphologies due to internal rotation and interactions with the external tidal field (Marklund et al. 2026), but remain limited in total mass and density due to the significant computational cost of direct N -body integrations. As such, they particularly resemble low-density GCs in M31 and the Milky Way. In total we use ten such simulations with an initial amount of 250k stars, three with 500k stars, and one with 1.5M stars.
- (ii) The second group refers to MOCCA simulations, and include four models of 47 Tucanae ($[\text{Fe}/\text{H}] = -0.78$ dex), one model for ω Centauri ($[\text{Fe}/\text{H}] = -1.5$ dex), and five additional high-density models ($[\text{Fe}/\text{H}] = -1.6$ dex) which are described in more detail in (Askar et al. 2025; Zhao et al. 2026, with two models from private communication Askar 2026). MOCCA simulations relax the computational constraints in the direct N -body method while also assuming spherical symmetry. As such, while they do not account for a morphological variety, they allow us to run simulations with considerably more stars, higher total masses, as well as higher densities. In total we use one such model with an initial amount of 1M stars, two with 1.5M stars, five with 2M stars, one with 2.4M stars, and one with 12M stars (roughly resembling ω Centauri). These models correspond to high-density and massive M31 and Galactic GCs.

These two groups of simulations therefore complement each other and ensure that the parameter space of our full dataset broadly matches that of real GCs in M31. Figure 1 shows the masses and ages of the mock images in our dataset (coloured points), together with the corresponding mass and age estimates for GCs in M31 from Usher et al. (2024). Although a few gaps remain, the dataset overall covers most of the relevant parameter space and the entire dynamical evolution of GCs from time 0 to ~ 14 Gyr.

Each mock image is created based on a snapshot from one of these simulations at different time steps. For the ROLLIN’ simulations, the frequency of the snapshots vary between 1 – 15 Myr

⁵ Note that we call these images backgrounds, but in principle they could actually be foreground images.

³ <http://dx.doi.org/10.17909/T91S30>

⁴ <https://reproject.readthedocs.io/>

depending on the simulation, whereas for the MOCCA simulations it is either 50 or 100 Myr, depending on the simulation and the age of the GC. As such, different simulations will be more or less prominent in the overall dataset, and in particular, there will be a bias towards the ROLLIN' simulations. This is also partly by construction as we are interested in exploring the morphology of GCs.

However, to mitigate biases within the dataset, we restrict the number of unique snapshots of simulations with 250k stars to $\min(N, 2000)$, while for the more massive ROLLIN' simulations (500k-1.5M stars initially) the total number of snapshots are set to $\min(N, 3000)$ where N is the total amount of available snapshots from a simulation. This leaves us with 37k unique ROLLIN' snapshots. On the contrary, we consider all available snapshots from the MOCCA simulations, amounting to a total of ~ 1.5 k unique snapshots.

To generate mock images from these simulated models of GCs that properly treat typical observational characteristics of HST M31 data, we have developed a forward modelling algorithm, CREMANT (Creating REalistic Mock imAges for Numerical simulaTions), inspired by previous codes such as COCOA⁶ (Askar et al. 2018) and SISCO (Bianchini et al. 2015; de Vita et al. 2017). The algorithm makes use of

- (i) the flexible stellar population synthesis (FSPS) code (Conroy et al. 2009; Conroy & Gunn 2010) to compute individual stars' magnitudes in the passbands we consider for our mock images. The stellar parameters from the simulations that are passed to FSPS are mass, radius, luminosity, and metallicity, and the effective temperature and surface gravity are calculated in FSPS. These stellar properties (excluding metallicity which is passed as an input to the simulations) are, in turn, provided by the stellar and binary evolution from the SSE and BSE prescriptions of Hurley et al. (2000, 2002) incorporated within the star cluster evolution codes (we refer to Kamlah et al. 2022, for the stellar evolution implementation in our simulations). Within FSPS, we use PADOVA isochrones (Bressan et al. 2012) and the BASEL spectral library (Lejeune et al. 1997, 1998; Westera et al. 2002).
- (ii) Two angles and a distance, as free parameters, that define geometrical projections on the plane of the sky. The field of view of our images ($12'' \times 12''$) and the pixelscale ($0.04''$) are also free parameters⁷ in the algorithm, and together define a two-dimensional pixel-grid onto which the particles from a simulation can be projected. To generate a projection, we first define a versor, which represents a rotation axis with respect to the coordinate axis of the simulation, and a rotation angle $\phi \in [0, \pi)$. Together, the versor and rotation angle determine the inclination⁸ of the cluster. The projected cluster can subsequently be rotated in the image plane by an additional angle $\theta \in [0, 2\pi)$, applied around the axis perpendicular to the projection plane. These two rotations fully specify the apparent orientation of the simulated cluster. The cluster is then placed at a chosen distance which, together with

the apparent orientation, determines the relative positions for each star within the pixel-grid.

- (iii) Point-spread functions (PSFs), derived from non-saturated bright stars in the outer regions of M31 (PHAT brick 22 with each passband aligned north-east) using DrizzlePac⁹. We include four distinct PSFs per passband, including STScI's empirical model¹⁰, all accessible via CREMANT. Multiple PSFs are included to prevent the network from memorising an imperfect PSF model with respect to the backgrounds' PSF. We oversample each mock image and PSF with a factor of four to allow for sub-pixel variations before convolving the images to produce the final mock image.

2.3. Modelling noise

Our mock cluster images are idealised source-only realisations, effectively corresponding to a noise-free limit in which the GC flux is known without photon-counting noise and with no background. To generate realistic observations we combine our mock images with backgrounds from real observations following the steps outlined in Bialopetravičius et al. (2019). We first transform the mock images into units of counts per second, matching the background images, and then:

- (i) compute the median pixel value of the background and add it to each pixel of a mock cluster image to place the source model on a realistic count scale;
- (ii) draw a Poisson realisation for each pixel in total counts, with the mean set to the corresponding pixel value;
- (iii) subtract the median offset and add the real background image;
- (iv) convert the pixel values to magnitudes (ST system).

This procedure accounts for both photon-counting noise and the spatial structure of the observed background, while preserving the source properties of the mock cluster. We do not apply any magnitude cuts, since the background already sets the effective observational baseline. During training (see Section 3.4), 10% of the mock images are also generated background-free to improve generalisation to real HST data with negligible contamination¹¹. Because CREMANT currently does not implement extinction, each background is corrected for Milky Way extinction using dustmaps¹² (Schlegel et al. 1998; Green 2018).

The noise modelling is carried out on the fly during training, effectively extending our total dataset by four orders of magnitude since each mock image can be paired with any of the 12264 backgrounds. A qualitative comparison between examples of our mock images and real GCs in M31 is shown in Figure A.1.

2.4. Building the datasets

For each snapshot of a simulation, we create a total of 6 mock images, with three different projection angles, and each angle at two different distances. In total, we create approximately 215k mock images of GCs from the ROLLIN' simulations, and roughly 9k images from the MOCCA simulations.

⁹ https://github.com/spacetelescope/hst_notebooks/tree/main/notebooks/DrizzlePac

¹⁰ <https://www.stsci.edu/hst/instrumentation/wfc3/data-analysis/psf>

¹¹ In such cases, we omit step (iii) above, and stars fainter than the typical HST background therefore remain visible.

¹² <https://dustmaps.readthedocs.io/en/latest/index.html>

⁶ <https://github.com/abs2k12/COCOA>

⁷ The pixel scale is set by the choice of telescope (HST), and the field of view is chosen to minimise image size while including as much of the GC as possible. In particular, we used the more massive, low-density and extended ROLLIN' models at the distance of M31 to set a reasonable FoV, and then cross-checked against observed M31 GCs, bearing in mind that their full spatial extent is not known exactly.

⁸ We consider the inclination with respect to the z-axis and the line of sight. The z-axis corresponds to the rotation axis for the ROLLIN' simulations.

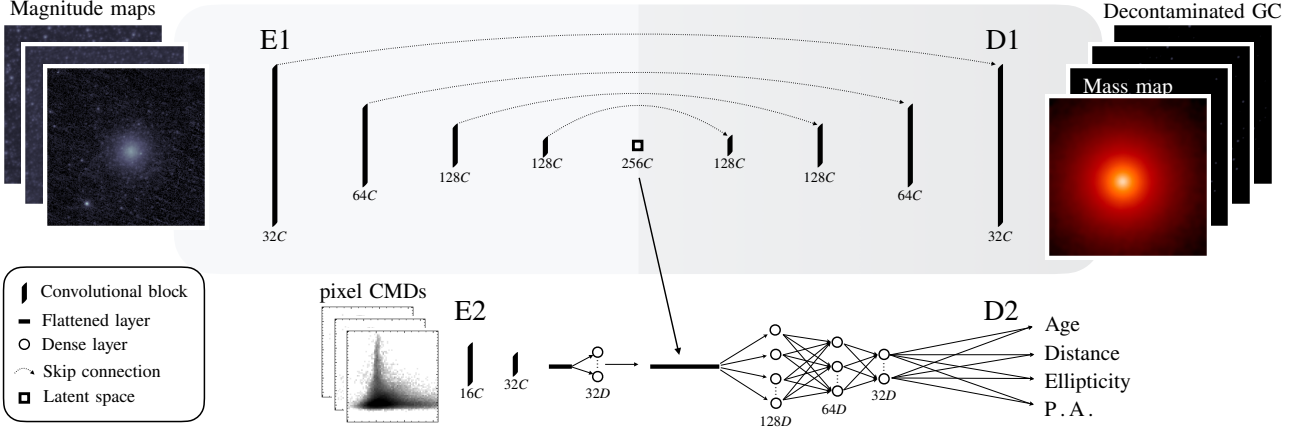


Fig. 2. Architecture of π -DOC. The different encoders are labeled E1-E2 and the respective decoders are labeled D1-D2. The corresponding number of filters (channels) are indicated below the blocks. Skip connections are indicated by dotted arrows. The regression head (D2) combines the latent features from each of the encoders, and the number of dense layers are indicated beneath the symbols.

The distances for the mock images are chosen such that they cover the expected range for GCs in M31. [Holland \(1998\)](#) put most of the GCs at similar distances to M31 itself, and from the Galactic population of GCs we know they may extend as far as ≥ 100 kpc from the Galactic Center (see e.g. [Harris 2010](#)). As such we choose distances to be sampled from a uniform distribution centred at the distance of M31, 785 ± 100 kpc (here using the distance estimate to M31 from [McConnachie et al. 2005](#)).

Some quantities are set directly by the simulations, such as the age (sampled uniformly in linear scale¹³), total luminosity, and total mass (see [Bianchini et al. 2026](#); [Marklund et al. 2026](#)). The projected images, the two-dimensional mass distribution, ellipticity, and the corresponding position angle (P.A.) of the minor axis¹⁴, are also set by the simulations, but further depend on the two geometrical projection angles defined in CREMANT (Section 2.2), as well as the distance. These angles and the versor are all sampled uniformly.

We compute a GC’s apparent ellipticity $e = 1 - b/a$ using the second-moment tensor method on the stellar distribution within the FoV, where a and b are the major and minor axes, respectively. This robust approach reliably traces the cluster’s intrinsic morphology ([Fréour et al. 2026](#); [Marklund et al. 2026](#)), facilitating clearer interpretation of our results. The position angle is measured east of north over $[0^\circ, 180^\circ)$.

Apart from the observational images, we also construct pixel colour-magnitude diagrams (CMDs) that are 64×64 pixels¹⁵. In cases where three (two) passbands are used, we get three (one) possible pixel CMDs: F475W vs F336W–F475W, F814W vs F336W–F814W, and F814W vs F475W–F814W.

We split the entire datasets into three parts: training ($N_{\text{train}} \sim 130\text{k}$), validation ($N_{\text{val}} \sim 44\text{k}$), and testing ($N_{\text{test}} \sim 50\text{k}$). The three sets are constructed in such a way that no mock image exists in more than one of the sets. This also applies to the real backgrounds, which in turn are distributed among the three sets in proportion to that set’s number of mock images (e.g., roughly 60% of the backgrounds are in the training dataset).

In order to get a diverse training dataset, it is customary to perform data augmentation. We combine the mock images and the real backgrounds on-the-fly when loading each batch, effectively obtaining a new set of images for every epoch. Each image is also rotated either 0° , 45° , 90° , or 135° .

3. Network

In this section we describe the new architecture of π -DOC as well as the corresponding inputs and outputs. π -DOC is a convolutional autoencoder based on the U-net architecture ([Ronneberger et al. 2015](#)), but it has been deeply modified to handle hybrid-inputs and to produce multi-modal outputs. The previous proof-of-concept version of π -DOC was split into two parts: an autoencoder that mapped the input image to a two-dimensional mass distribution, and a classical convolutional classifier that predicted the cluster’s age and distance from the same input. This separation is inefficient, since both parts must be trained independently, and precludes a joint latent representation — which could otherwise be useful given that certain outputs, such as age and the mass distribution, are inherently related ([Baumgardt 2017](#); [Bianchini et al. 2017](#)).

For the new architecture we have combined the two networks into one multi-task network. In particular, it is now capable of handling multi-band photometric images and pixel CMDs. Inputs are described in more detail in Section 3.1. We note that the inclusion of pixel CMDs was found to significantly improve age and distance estimates in early testing of the new network architecture, in particular due to a more explicit encoding of the shape of the associated isochrones. There are also three completely new outputs: a decontamination of the input multi-band photometric image (e.g. removal of the background), and the GCs’ ellipticity and corresponding position angle for the minor

¹³ The distribution is only approximately uniform, as the largest N -body simulation was evolved only up to 12 Gyr, three MOCCA simulations were evolved up to 15 Gyr, the ω Cen equivalent model is additionally under-sampled at all evolutionary stages due to the computational cost of its magnitude calculations, and early snapshots are entirely absent, while all the remaining simulations were run up to 14 Gyr.

¹⁴ In projection, the minor-axis position angle is equivalent to the major-axis angle shifted by 90° . We adopt the minor axis here as a matter of convention and for consistency with prior studies (e.g. [Fréour et al. 2026](#); [Marklund et al. 2026](#)).

¹⁵ We note that using pixel CMDs with an increased resolution (128×128 , 256×256) did not improve performance.

axis, see Figure 2. We describe all outputs in more detail in Section 3.2.

The additional inputs and outputs have also resulted in a substantial re-work of π -DOC’s overall design, and we describe the corresponding architecture in more detail in Section 3.3. Additionally, because the PHAST survey is available only in two passbands, we construct the network for both two and three passbands, and we denote them π -DOC₂ and π -DOC₃, respectively.

3.1. Input data

The main inputs to the network consist of the multi-band photometric images alongside pixel CMDs. π -DOC₂ uses two magnitude maps and one CMD, and π -DOC₃ uses three magnitude maps and three CMDs. We also experimented by explicitly providing colour maps as part of the inputs, but did not find this to yield any improvements. This is reassuring because it suggests that this information is already encoded in the raw multi-band images. See Section 2 for more details on the construction of the input data.

All inputs are standardised, though not uniformly. The magnitude maps across all passbands are standardised jointly — using a single combined mean subtracted from every band — rather than independently per band, and without enforcing a unit standard deviation. This preserves the relationship between magnitudes and colours, so that even after standardisation, the difference between two bands still corresponds directly to their colour. The pixel-wise CMD number counts, by contrast, are standardised to zero mean and unit standard deviation. All standardisation constants are computed once by iterating over the training data (e.g. one epoch, with each mock image combined with a background; see Section 2.3). Since noise augmentation is applied on-the-fly, no epoch sees exactly the same images; however, given the size of the training set ($\sim 130k$ images), the standardisation constants computed from a single epoch are statistically stable and not meaningfully affected by this variation.

3.2. Output data

π -DOC is tasked to decontaminate the input images in all passbands from the background, map the corresponding two-dimensional mass distribution within the FoV (same pixel-scale as the input images), and predict the cluster’s age, distance, ellipticity, and position angle for the minor axis (P.A.). Instead of predicting P.A. directly, the network predicts $\cos(2P.A.)$, $\sin(2P.A.)$; the resulting P.A. is then calculated using the two-argument arctangent (atan2) as $\frac{1}{2} \text{atan2}(\sin(2P.A.), \cos(2P.A.))$. This approach removes large errors at the periodic boundary where 0° and 180° are degenerate.

We also apply a Gaussian smoothing kernel to the mass maps ($\sigma = 4$). This is to avoid the network having to deal with predicting highly pixelated images, especially in the outskirts of the clusters where the stellar counts are sparse. Note that, following the proof-of-concept version of π -DOC (Chardin & Bianchini 2021), we have tested applying such a smoothing on the input and output magnitude maps as well, but this did not improve performance in our setup. The input images and the decontaminated output images therefore remain consistent with HST’s pixel scale.

All of these outputs are standardised during training except for the position angle since we infer $\sin(2P.A.)$, $\cos(2P.A.)$. The decontaminated magnitude maps are standardised to be consis-

tent with the input magnitude images (see Section 3.1), and the remaining outputs are standardised to have a mean of zero and a standard deviation of one.

3.3. Network architecture

The inclusion of additional input data, such as the CMDs, enables more complex operations within the network, encouraging it to extract both complementary and distinct features from multiple representations of the same underlying system. The new architecture reflects this, and we show a schematic overview in Figure 2. There are four different components, which can be split into two encoding parts (E1 and E2) and two decoding parts (D1 and D2)¹⁶:

- E1 is an encoder applied on input magnitude maps (first row of Figure 2) that consists of five convolutional blocks.
- E2 is a feature extractor applied on CMDs (second row of Figure 2). The convolutional blocks are slightly different from those employed in E1. The flattened representation is projected to a 32-dimensional feature vector before concatenation with the larger latent space, increasing its dimensionality and allowing it to contribute more information to the subsequent predictions.
- D1 is a decoder (first row of Figure 2) consisting of five deconvolutional blocks and decontaminates the input magnitude maps.
- D2 is the regression head in the second row that predicts the scalar outputs (age, distance, ellipticity, and position angle) from the concatenated and flattened latent spaces of the two encoders E1 and E2.

Finally, to reduce overfitting we use dropout and set it to 0.1 for each version.

3.4. Training

For all versions of π -DOC, the training is governed by the same loss function, which we take to be a weighted sum of each output’s mean-square-error (MSE). Note that we also experimented with incorporating a perceptual loss function for predicting the mass and luminosity, as per Chardin & Bianchini (2021), but we did not observe it to yield any improvement. As a result, we consider the total loss function as

$$\mathcal{L} = \sum_{i=1}^I w_i \|y_i - \hat{y}_i\|^2, \quad (1)$$

where I denotes the number of outputs, each y_i denotes one output: the decontaminated magnitude maps which is considered as a singular output regardless of the number of passbands, mass map, age, distance, ellipticity, and position angle. Consequently, $\hat{y}_i$ is its estimate by the network. w_i denotes a scaling factor that is set to ensure each output MSE is in the same range. The weights are set to one for all outputs except for the magnitude and $\cos(2P.A.)$, $\sin(2P.A.)$, which are set to 0.2, 0.5, and 0.5, respectively. Furthermore, for GCs with $e = 0$ we dynamically set a zero-weight associated to the P.A. since it is not defined.

Both networks have been trained on 4 NVIDIA H200 GPUs. We use the ADAM optimizer (Kingma & Ba 2014), and a learning rate of $\eta = 10^{-4}$. Moreover, we implemented a learning-rate

¹⁶ The latter of these parts illustrate a feature extractor and a regression head rather than an encoder and decoder, but for simplicity we denote them as E2 and D2.

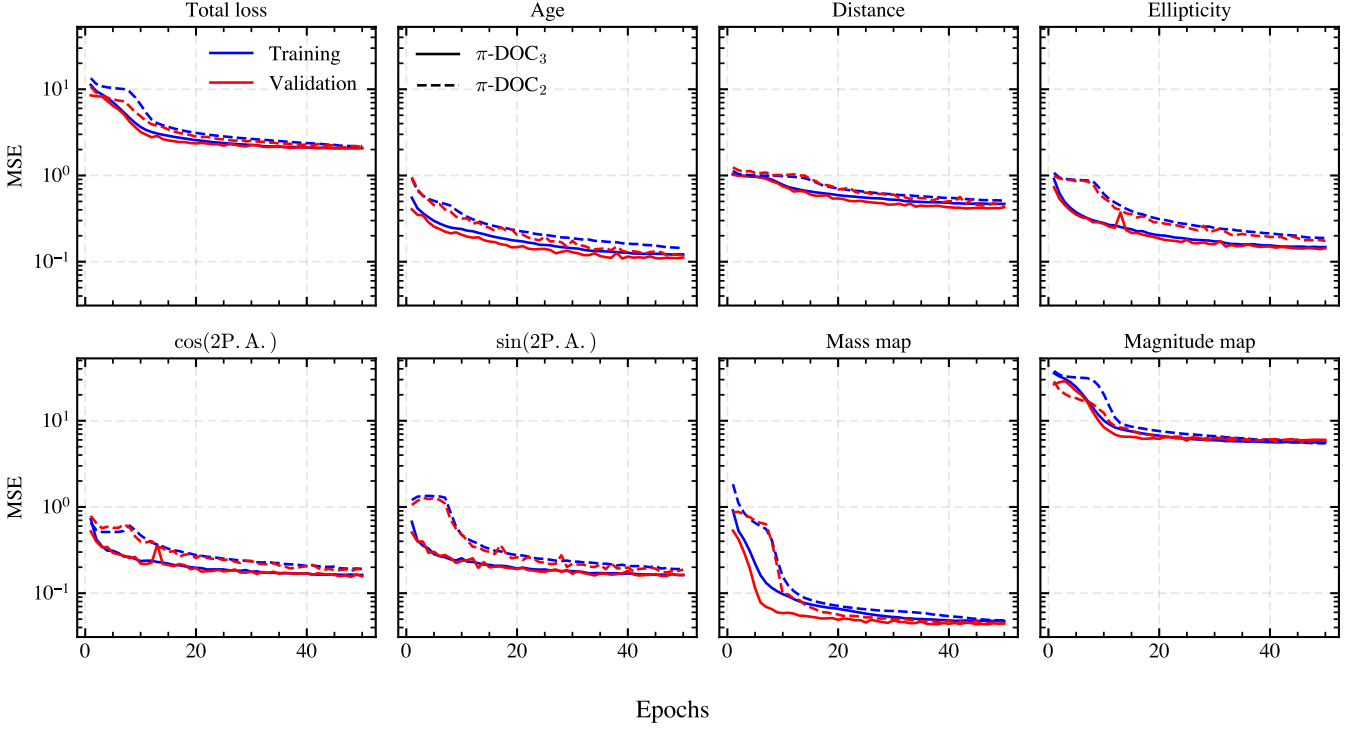


Fig. 3. Training curves for the two π -DOC variants considered in this study. Blue (red) lines correspond to the training (validation) MSE per output across epochs. The three-passband network converges faster and achieves lower overall MSE than its two-passband counterpart, although this difference is not substantial. We interpret this as a result of the larger (but not deeper) network capacity.

Table 1. MSE of π -DOC₂ and π -DOC₃ applied on the validation data set for the different outputs.

MSE	π -DOC ₂	π -DOC ₃
Magnitude map	5.86	6.00
Mass map	0.05	0.04
Age	0.12	0.11
Distance	0.44	0.43
Ellipticity	0.17	0.14
Position angle	0.19	0.16
Total weighted	2.17	2.09

Notes. MSE for each quantity. The MSE for the P.A. is taken to be the average between $\sin(2P.A.)$ and $\cos(2P.A.)$.

scheduler halving after 2 epochs without validation improvement and an early stopping after 5 epochs. Figure 3 shows the training and validation performance over epochs for each network. Training ran for a maximum of 50 epochs to save computational time with no early stopping triggered, yielding converged models with stable validation performance (total training required ~ 1200 GPU hours for each model).

We indicate the total MSE, and the output-specific MSEs, for the validation dataset for each version of π -DOC in Table 1. We find that the inclusion of F336W marginally improves performance for all outputs except the decontaminated magnitude map. We attribute this to two competing effects: the additional passband provides more constraining information, but also increases the dimensionality of the decontamination task, which must isolate the GC signal from the field and background consistently across all bands. For decontamination, this added complexity appears to outweigh the benefit of the extra information.

4. π -DOC's performance on mock images

4.1. Illustrative examples

To assess the network's performance on mock observations, we first consider four examples from the test set; unless stated otherwise, all predictions throughout this section are obtained with π -DOC₃. Figure 4 shows the corresponding pseudo-RGB images constructed from the three passbands.

Throughout this work, we quantify the level of background contamination using the difference between the integrated GC and background magnitudes in the F814W passband, hereafter denoted S/B. Panels (a) and (b) have negative S/B values (-0.76 and -0.61 , respectively), indicating that the integrated cluster light exceeds the background, whereas panels (c) and (d) have positive S/B values (3.0 and 2.5 , respectively), corresponding to increasingly background-dominated observations.

The four examples were selected to span a range of cluster properties, background contamination levels, and simulation types: (a) a relatively young, low-density ROLLIN' cluster; (b) an old, dense MOCCA cluster; (c) an older, fainter ROLLIN' cluster; and (d) an old but relatively dense MOCCA cluster. The second and third rows show the network's decontamination output¹⁷, together with the scalar predictions for the distance and age, as well as the total luminosity and half-light radius derived from the decontaminated map.

Across the four examples, the decontamination and associated scalar outputs remain close to the ground truth. In the negative S/B examples individual stars are retained, though their colour is not guaranteed to be correctly inferred. Instead, there

¹⁷ For visualisation purposes, the pixel values are restricted to magnitudes between 25.7 and 31. The underlying π -DOC predictions are not limited to this range.

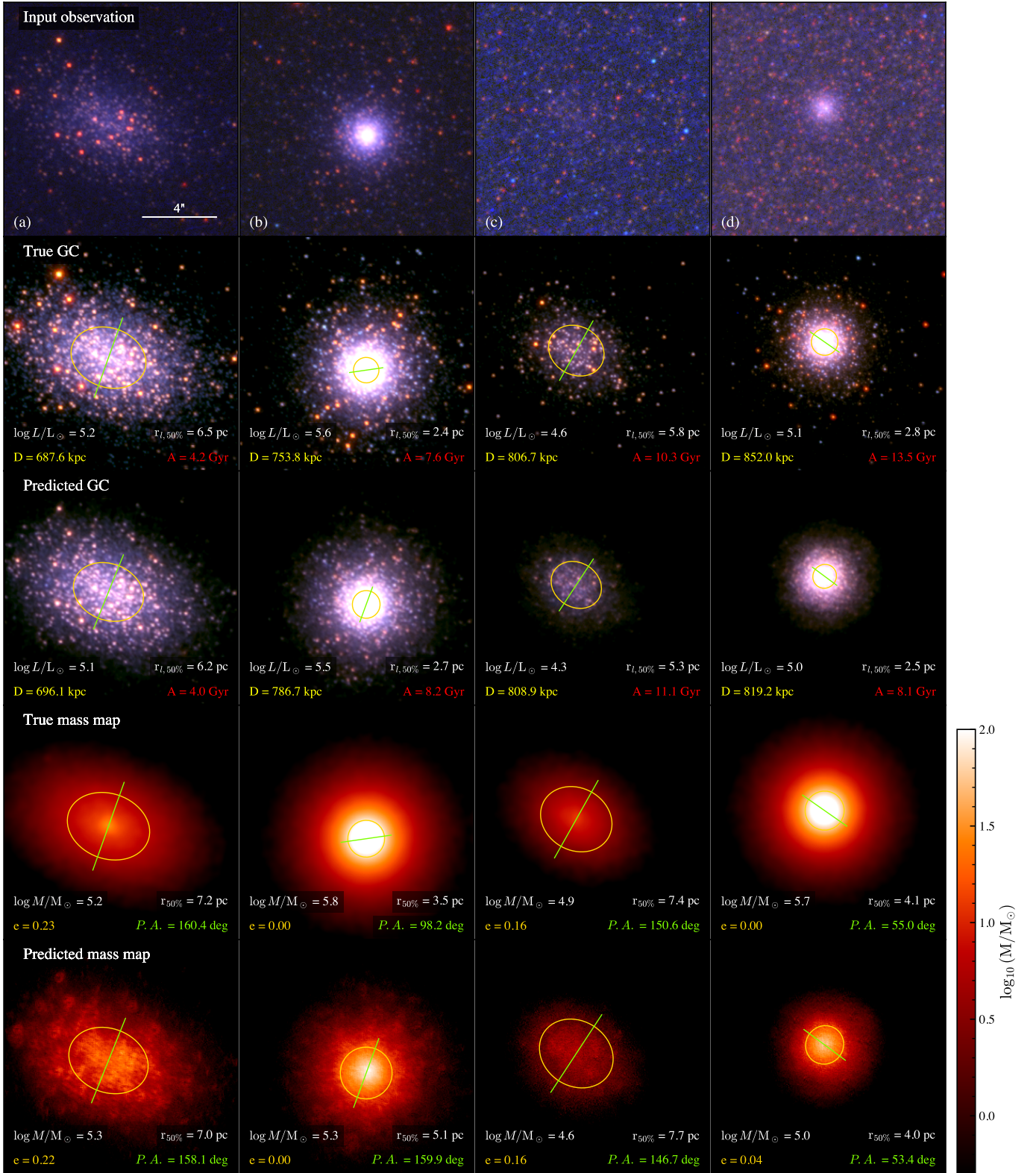


Fig. 4. Mosaic example of four representative mock images (ROLLIN' models in panels a and c, and MOCCA models in panels b and d) in both GC- and background-dominated regimes (top row), the true decontaminated cluster (second row), the predicted decontaminated cluster (third row), the true mass distribution (fourth row), and the predicted mass distribution (bottom row). The decontaminated panels also indicate the true and predicted values for total luminosity and half-light radius ($r_{l,50\%}$) within the FoV, distance, and age. Similarly, the mass distribution panels indicate the total mass and half-mass radius ($r_{50\%}$) within the FoV, the ellipticity, and the position angle. The golden ellipse in each panels indicate the ellipticity and has its semi-major axis placed at the half-light or half-mass radius, whereas the green line indicates the position angle of the minor axis. The true and predicted decontaminated cluster images are scaled differently to the input observations in order to highlight the differences in decontamination across the full extent of the cluster.

is a tendency to assign similar magnitudes per pixel across the various passbands with the resulting GC appearing whiter. The successful decontamination of the clusters in the brighter backgrounds (panels b and c), demonstrates that π -DOC can distinguish the overall diffuse distribution of the GC from the background even in challenging noise regimes. On the other hand, not all stars (or pixels) are successfully recovered — in particular, those farther from the cluster centre are more likely to be missed. This, however, does not appear to affect global quantities significantly.

The remaining two rows show the true and predicted 2D mass distributions, and the corresponding total mass and half-mass radius (derived directly from the mass maps). The total mass is constrained to within a factor of two for the two ROLLIN' models in panels (a) and (c), while the MOCCA models in (b) and (d) amount to an underestimation by a factor of ~ 3 and ~ 5 , respectively. This is an indication that the network struggles to generalise between our ROLLIN' and MOCCA models.

As a matter of fact, our training data are dominated by N -body models, so high-mass clusters are primarily represented by the comparatively small number of MOCCA models included in the training set. These results may therefore partly reflect π -DOC's limited ability to generalise in this poorly sampled regime, rather than a genuine astrophysical trend. An alternative, and not mutually exclusive, explanation is the retention and spatial distribution of non-visible mass within the GCs (e.g. stellar remnants). Example (b) hosts both an intermediate-mass black hole ($M \sim 6000 M_\odot$) and a substantial white dwarf (WD) population ($M \sim 1.3 \times 10^5 M_\odot$), together accounting for roughly 30% of the missing mass, while the non-visible remnant population in example (d) accounts for 25% ($M \sim 1 \times 10^5 M_\odot$) — implying remnant mass fractions on the order of 15-25% in the MOCCA models. We find a similar fraction for the ROLLIN' models, but their spatial distributions differ substantially: in the N -body examples, WDs closely trace the overall mass distribution (the ratio of Lagrangian radii are $r_{10,WD}/r_{10,all} \sim 1$, $r_{30,WD}/r_{30,all} \sim 0.9$, and $r_{50,WD}/r_{50,all} \sim 0.84$ - 0.88)¹⁸, whereas in the MOCCA examples the WDs are considerably more centrally concentrated ($r_{10,WD}/r_{10,all} \sim 0.8$, $r_{30,WD}/r_{30,all} \sim 0.7$, and $r_{50,WD}/r_{50,all} \sim 0.65$ – 0.74).

We suggest this may contribute to π -DOC's underperformance on MOCCA-like clusters. Since the network infers total mass from the visible light distribution alone, it must implicitly learn a visible-to-total mass (or alternatively a spatially dependent M/L ratio) mapping calibrated mainly on the N -body-dominated training set, where remnants are not strongly concentrated relative to the total mass. Applied to MOCCA-like clusters, where a comparable remnant fraction is instead concentrated within a much smaller region, this mapping would be expected to undercorrect the innermost pixels, while the greater compactness of MOCCA clusters leaves less room to compensate. We explore these possibilities in more detail in Section 7.

The two MOCCA clusters in panels (b) and (d) are intrinsically spherical, and therefore have zero ellipticity. The ellipticity is correctly recovered for panel (b), while π -DOC predicts a small non-zero value of 0.04 for panel (d). This offset may provide a rough sense of the absolute precision achievable by the model in background-dominated regimes. The position angle is formally undefined for both clusters; the reported values are included only for completeness, and the close agreement in panel (d) is therefore coincidental. In contrast, both the ellipticity and

position angle of the two ROLLIN' models, which are characterised by axisymmetry, are recovered.

Although these four mock images represent only a small subset of our full test set, they illustrate how π -DOC performs under a variety of realistic observational conditions. The network successfully decontaminates the cluster images across the range of background levels considered, although the scalar predictions degrade for the most strongly background-dominated cases. In particular, example (d), with $S/B = 2.5$, shows the largest age discrepancy (~ 5 Gyr compared with $\lesssim 0.8$ Gyr for the other examples) and a substantial mass offset. Example (c) also shows a noticeable mass offset. The distance errors are ~ 30 kpc for the two MOCCA models, compared with $\lesssim 10$ kpc for the two ROLLIN' models. Within each simulation family, the fainter examples (c) and (d) show larger discrepancies than their brighter counterparts (a) and (b), respectively. These examples therefore suggest that both background contamination and differences between the ROLLIN' and MOCCA models may influence the accuracy of the scalar predictions.

4.2. Performance on test set

Qualitatively, the performance shown in Figure 4 is representative of the full test sample. Figure 5 shows the prediction distributions for all π -DOC outputs on the test set, alongside the corresponding training and test parameter distributions. The training data span the full parameter space of the test set, and the predicted distributions closely track the test distributions across this range, indicating the broad applicability of π -DOC.

Panels (a-c) show the decontamination performance in the three photometric passbands, that is, the total integrated magnitude of the GC in each FoV. The integrated magnitudes are slightly overestimated across the entire range (e.g. their brightness is underestimated), and this trend is more severe for the faintest clusters (see the tail at large magnitudes), which is expected since faint clusters are harder to distinguish from the background.

The total mass distribution in panel (d) shows two distinct peaks in the test set, while the network instead predicts a more unimodal distribution and underestimates both the least massive clusters and the most massive ones. This largely follows from the decontamination results above: clusters whose brightness is underestimated typically also have their total mass underestimated. At the massive end, this may also reflect a bias toward lower mass predictions, likely affected by the relatively poor representation of massive simulations in the training data noted above.

The training and test distributions are uniform for both age (panel e) and distance (panel f). The predicted age distribution is also fairly uniform, aside from two modest peaks near 1–2 Gyr and 9 Gyr. Distance predictions, by contrast, cluster toward the mean of the test distribution. This likely reflects the intrinsic difficulty of constraining distance, but also mirrors a well-known behaviour in neural networks (Kang et al. 2024), which tend to perform less reliably near the edges of the training domain: because the network implicitly learns the bounds of the parameter space, it avoids extrapolating beyond them and instead biases predictions toward interior values.

Predictions of ellipticity (panel g) and position angle (panel h) closely match the ground truth across the full range. Small ellipticities ($\lesssim 0.1$) are overestimated, likely reflecting a bias against predicting exactly zero for the MOCCA models (see e.g. Figure 4). At these small ellipticities, the position angle is either undefined ($e = 0$) or poorly defined, since there is no clear major

¹⁸ These are the radii within which 10%, 30%, and 50%, of the mass is confined.

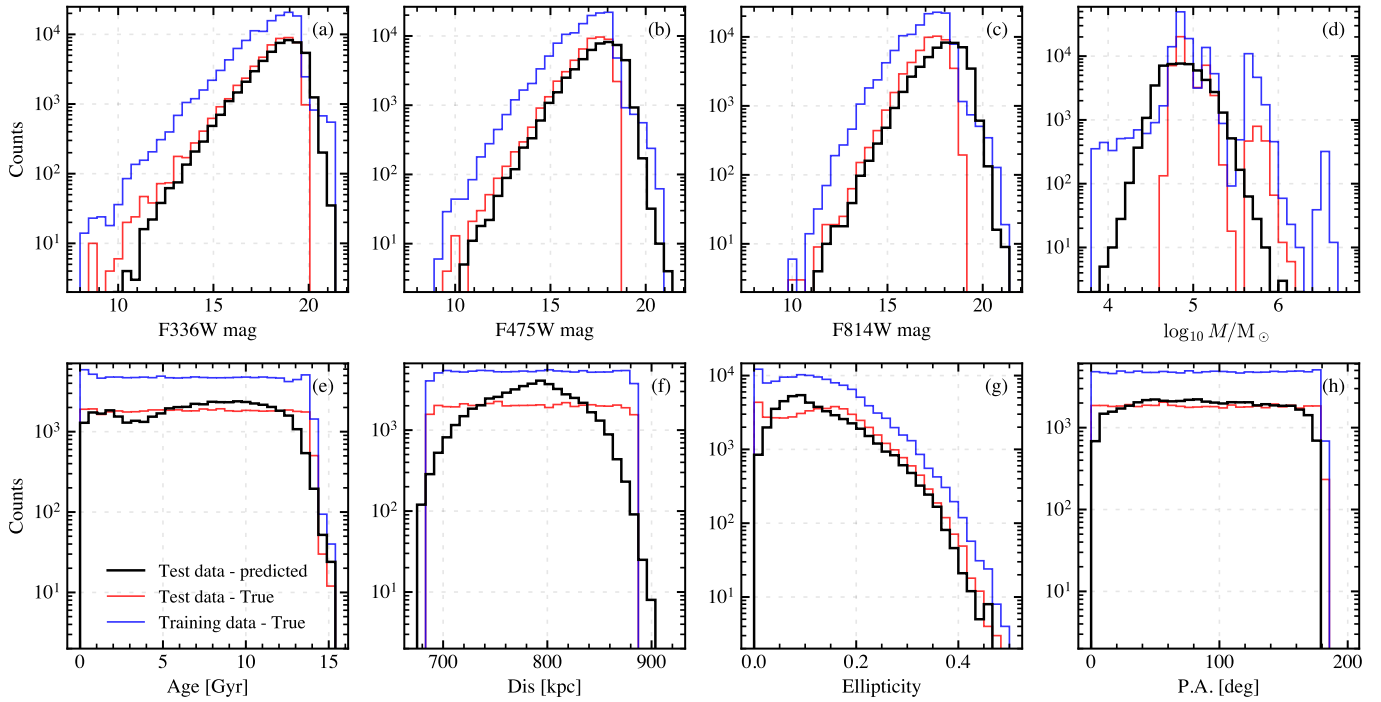


Fig. 5. Distribution of the predictions for the main outputs of π -DOC (black lines), including the corresponding distributions for the test data (red lines; ground truths to predictions) and the training data (blue lines). When the distribution of the predicted test data aligns with the distribution for the corresponding ground truth, it implies that the network’s outputs are statistically unbiased as an ensemble, even though individual predictions may still deviate from their true values. Top row: (a) integrated FoV F336W magnitudes, (b) integrated FoV F475W magnitudes, (c) integrated FoV F814W magnitudes, (d) total mass in log-scale. Bottom row: (e) age, (f) distance, (g) ellipticity and, (h) position angle.

or minor axis; large position-angle errors are therefore expected in this regime and are not to be considered a concern.

Overall, π -DOC₂ and π -DOC₃ agree well across all predicted quantities, with the former showing larger absolute uncertainties. For an explicit comparison between the two models, see Appendix B.

4.3. Biases within the network

The previous section explored how the network retains the overall distributions of the test set. Here, we explicitly aim to address the cases where the predictions are most uncertain. Figure 6 shows a direct comparison of the differences between the predictions and the ground truth of the test data, as well as the corresponding correlations between the quantities. We refer to the difference between a predicted quantity and its ground truth as the error. To facilitate the discussion below we keep the ground truth magnitude for the F336W passband in column (a) rather than the corresponding error. The inset table indicates the median, 1σ , and 2σ , of each quantity’s 1D error distribution.

Column (a) shows that the typical errors increase for most quantities the fainter the clusters are. As previously mentioned, this likely reflects an increased difficulty in distinguishing the GC from the background. Indeed, the majority of the GCs in our data set are relatively faint (see also Figure 5), having an average integrated magnitude of 18.52, 17.46, 17.70 for F336W, F475W, and F814W, respectively. In comparison, the average integrated magnitudes of the added backgrounds for the three passbands are 16.51, 15.20, 15.28, respectively. Therefore, most of the mock images have $S/B > 0$.

For brighter clusters (~ 12 – 17 mag), the errors are smaller, especially for age, ellipticity, and position angle, while the dis-

tance errors are roughly independent of brightness. In contrast, the total mass is typically overestimated for brighter GCs while the decontamination in F475W and F814W shows a nearly constant positive magnitude bias, with mean offsets of 0.43 and 0.64 mag, respectively. These discrepancies become more severe for both really faint and really bright clusters, but is nonetheless present across the entire range. There is a similar shift of 0.28 mag for the F336W passband. The overall uncertainty thereby increases with redder passbands: F336W is best constrained and F814W the worst. Because the brightness in each passband is underestimated by different amounts, it also means that the colours of the decontaminated GC are biased, with π -DOC predicting bluer colours on average.

These intrinsic errors are also comparable to limitations in the literature, and we regard the 2σ limits of the distributions in Figure 6 as a conservative confidence range for π -DOC’s predictions. For the F336W, F475W, and F814W passbands, the spread of the decontamination errors are confined to within ± 0.85 mag, but suffers from the aforementioned biases. The mass ($^{+0.30}_{-0.43}$ dex) and age ($^{+2.73}_{-3.57}$ Gyr) are well constrained compared to typical uncertainties for M31 GCs (Fan et al. 2010; Chen et al. 2016; Usher et al. 2024), and the mass confidence limit is only around a factor 1.5 wider than the typical uncertainties for fully resolved Galactic GCs (see e.g. Bellini et al. 2017; Zocchi et al. 2017). The lower limit of the mass errors is further exacerbated by the MOCCA models, which are consistently underestimated (see the separate island in the bottom part of the corresponding panel in column a). Moreover, a handful of outliers lie beyond the 2σ boundary, particularly for the brightest clusters where masses are underestimated by factors exceeding four; these rare cases mainly correspond to early snapshots ($\lesssim 200$ Myr) from the un-

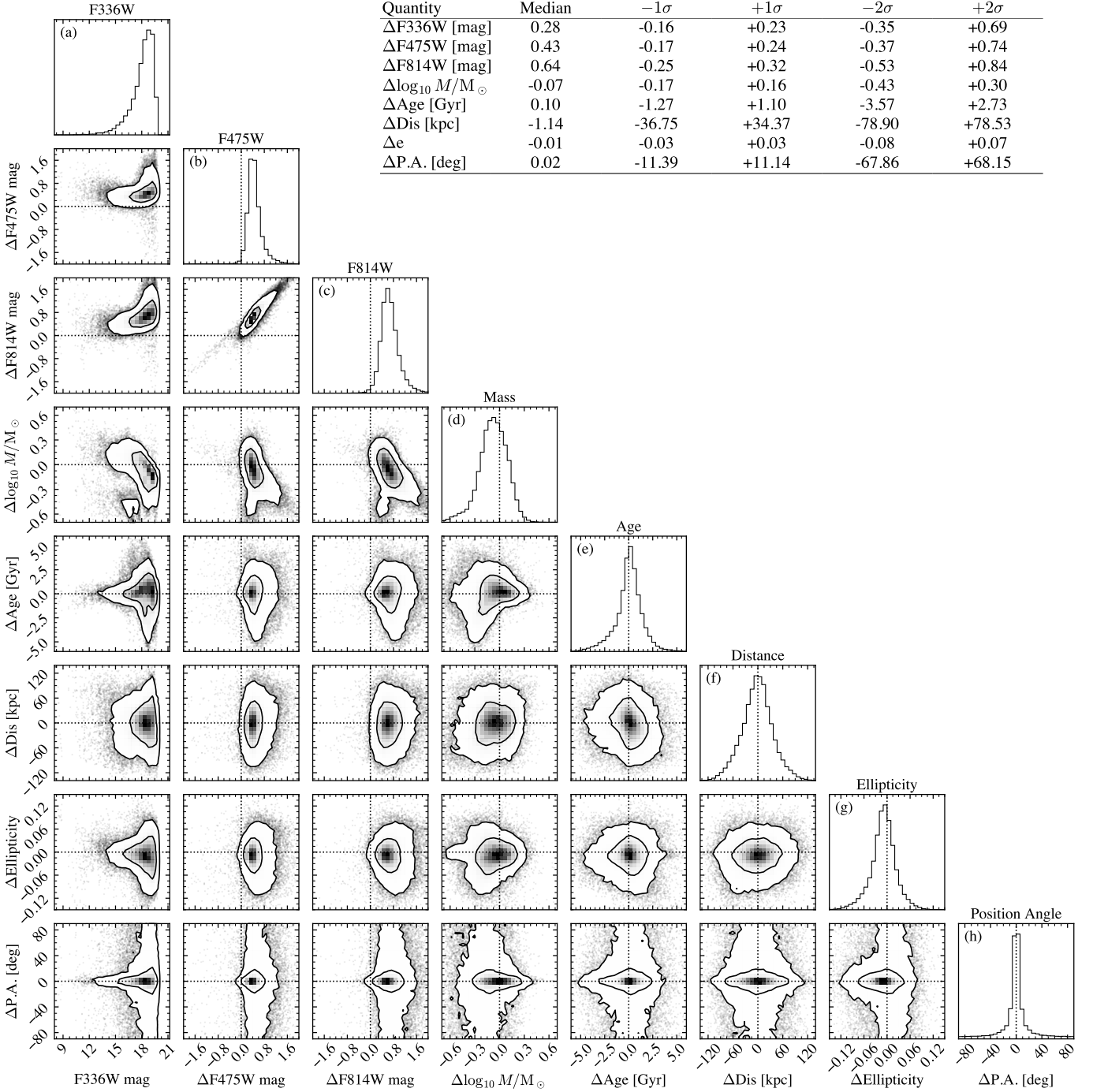


Fig. 6. Corner plot showing the distributions and correlated distribution of the errors of the main predicted outputs of π -DOC. The contours are set to σ , 2σ , and the dotted lines indicate the 0-axes (e.g. no discrepancy), where applicable. The diagonal shows the 1D error distribution for the (a) total and true total F336W magnitude of the GC within the FoV, (b) error for the total F475W magnitude of the GC within the FoV, (c) error for the total F814W magnitude of the GC within the FoV, (d) error for the total mass within the FoV in logarithmic scale, (e) error for the age in gigayears, (f) error for the distance in kiloparsec, (g) error for ellipticity, and finally (h) error for the P.A. in degrees. The inset table in the top right corner indicates the median, $\pm\sigma$, and $\pm 2\sigma$, values for each quantity's 1D distribution (e.g. panels along the diagonal).

derrepresented high-mass regime ($\gtrsim 10^6 M_{\odot}$) of the MOCCA simulations.

Distance ($^{+79}_{-79}$ kpc), ellipticity ($^{+0.07}_{-0.08}$), and particularly P.A. ($^{+68^{\circ}}_{-68^{\circ}}$) show larger uncertainties. The confidence range for the ellipticity and P.A. are primarily exacerbated by faint, background-dominated clusters which is evident from the elongated contours in column (a) of Figure 6. Excluding clusters fainter than 18 mag (F336W) and with an ellipticity below 0.05 (where the P.A. is

rather ill-defined) yields a more robust 2σ confidence range for the P.A. ($^{+22^{\circ}}_{-21^{\circ}}$).

We also include a similar exploration of biases and uncertainties with respect to the relative GC-to-background brightness that explicitly targets the S/B in Appendix C. Overall, we find a similar dependence on S/B as we do for the GC's brightness (see Figure C.1). Moreover, π -DOC demonstrates an impressive resilience to increasingly background-dominated images (see Fig-

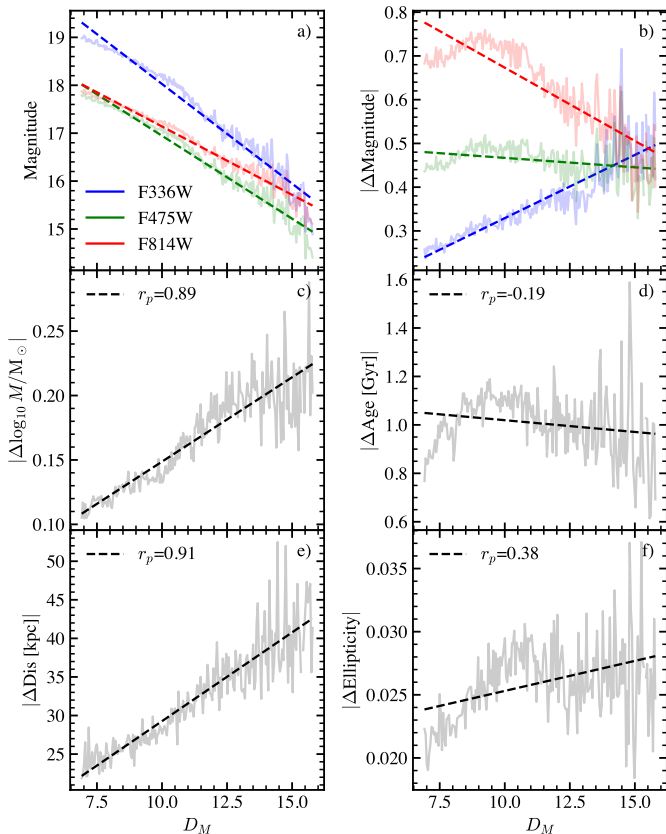


Fig. 7. Dependence on D_M for the total integrated magnitudes (panel a) with corresponding errors in panel (b), errors for total mass (panel c), errors for age (panel d), error for distance (panel e), and error for ellipticity (panel f). We find a strong linear correlation between D_M and the quantities in left-hand side column, while those in the right-hand side column show non-linear and weaker correlations.

ure C.2). Although this is beyond the purposes of this study, these results suggest π -DOC could be adapted to simultaneously detect GCs (Barbisan et al. 2022; Dold & Fahrion 2022; Singlow et al. 2022) in addition to inferring their properties.

5. Reliability of predictions and representational similarity

In this section, we consider two complementary approaches for assessing the reliability of π -DOC’s predictions. The first estimates the network’s own confidence in a given prediction (Section 5.1); the second assesses how representative an input is of the training data itself, via the Mahalanobis distance (Section 5.2).

5.1. Epistemic uncertainty

Estimating a network’s uncertainty for individual predictions in a reliable manner remains challenging. To quantify a predictive uncertainty in our model, we adopt Monte Carlo (MC) dropout, a computationally efficient approximation to Bayesian inference (see Gal & Ghahramani 2015a,b; Gal et al. 2017). During training, dropout layers randomly deactivate subsets of neurons to prevent overfitting. In MC dropout, this stochastic behavior is retained during inference by performing multiple forward passes with dropout enabled. Each forward pass samples a dif-

ferent realisation of the network’s weights, effectively drawing from an approximate posterior distribution over models. For a given input, we obtain an ensemble of predictions that provide a measure of the model’s so-called epistemic uncertainty.

For the majority of estimates, the epistemic uncertainties remain small and do not reflect the differences of the predictions to the ground truth. These values should not be regarded as physical uncertainties, but rather as the intrinsic confidence level of π -DOC, and we include them in our final catalogue and in the forthcoming sections. More details on our implementation can be found in Appendix D.

5.2. Mahalanobis distance

π -DOC is trained exclusively on synthetic data generated under controlled physical assumptions, therefore, when applied to real astronomical observations, there is no guarantee that all inputs lie within the same statistical distribution as the training set. Predictions made on inputs that differ significantly from the training distribution may therefore be unreliable, even if the model reports low internal uncertainty. A common approach is to define an out-of-distribution classification by thresholding the Mahalanobis distance (D_M) between a target’s latent-space representation and the training distribution, typically calibrated to a fixed false-positive rate on the training sample itself (Lee et al. 2018; Gonzalez et al. 2022). This binary classification is attractive for its simplicity, but it is inherently arbitrary: the training distribution has no natural density edge.

We therefore treat D_M as a continuous indicator of representational similarity to the training data, rather than as a binary classification based on an arbitrary threshold, and ask whether D_M traces the estimate quality: if the latent space organises clusters in a physically meaningful way, clusters further from the training manifold should show systematically larger discrepancies from the ground truth.

We extract intermediate feature representations from the joint latent spaces of E1 and E2 in Figure 2, keeping the features agnostic to specific output heads. These feature vectors are projected onto a lower-dimensional subspace¹⁹ using incremental principal component analysis (see e.g. Abdi & Williams 2010) to improve numerical stability and reduce redundancy. Assuming that the feature distribution of the training data can be approximated by a multivariate Gaussian in this low-dimensional space, we characterise it by its mean vector and covariance matrix. Given a new input, we compute D_M between its feature representation and the training distribution.

We find that D_M correlates strongly and linearly with the intrinsic brightness of the GC (Figure 7, with Pearson coefficients of $r_p \sim -0.9$ to -0.99 across passbands), indicating that the dominant axis of variation captured by this representation is closely tied to GC brightness. D_M also correlates with the brightness of the background and with the relative GC-to-background contrast, though these relationships are non-linear, exhibiting local extrema rather than a monotonic trend; we do not interpret these secondary trends further, as they likely reflect the combined influence of several correlated observational properties (GC brightness, background brightness, S/B, and plausibly other physical properties).

This provides a natural, if partial, explanation for several of the correlations seen in the remaining panels of Figure 7, which

¹⁹ We use a dimensionality of 256 for the lower-dimensional subspace. We also tested dimensionalities of 64 and 128, finding no significant differences in the resulting feature representations.

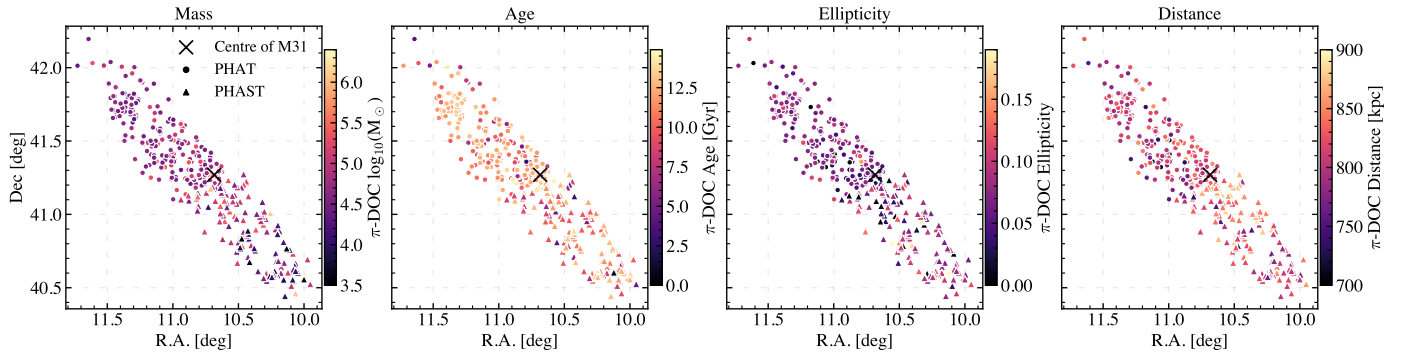


Fig. 9. GCs within the footprints of the PHAT (circles) and PHAST (triangles) surveys, colour-coded by π -DOC estimates of total mass (a), age (b), ellipticity (c), and distance (d). There is a tentative concentration of higher-mass GCs toward the centre, although this trend should be interpreted with caution. The oldest GCs according to π -DOC lie in the north-east region of M31, which also corresponds to larger inferred distances; this trend lacks a clear physical explanation and likely reflects biases in π -DOC and observational systematics such as extinction and age–metallicity gradients. Most GCs appear nearly circular, with the highest ellipticities typically found at larger galactocentric distances, likely reflecting a dependence on the S/B.

6. Application to M31 GCs

In this section we infer properties for 349 M31 GCs using π -DOC. Following the outline of Section 4, we begin by demonstrating π -DOC’s performance on four example GCs in Section 6.1 and compare their inferred values to independent studies. In Section 6.2 we apply π -DOC on all GCs within the PHAT and PHAST footprints, using π -DOC₃ and π -DOC₂, respectively. We then compare our findings to four independent studies in Section 6.3. The inference speed of π -DOC is ~ 10 GCs s^{−1}, so the full M31 dataset of 349 GCs required only about half a minute²⁰. We provide an online repository containing decontaminated images, mass maps, and all remaining predicted quantities, including epistemic uncertainties and D_M (for a reliability and similarity-score to the training data) for all GCs considered in the PHAT and PHAST surveys. A subset of the full catalogue are also reported in Table E.1; the complete results are available in the online supplementary material.

6.1. Mosaic example of four GCs

The four examples shown in Figure 8 have been chosen based on their D_M to the training manifold: (a) B386–G322 has the smallest D_M , (b) B186 lies at the 20th quantile, (c) B206 lies at the 40th quantile, and (d) B371–G303 lies at the 80th quantile. Each GC is located in the PHAT survey, and hence we employ π -DOC₃. Despite the four clusters representing examples of decreasing similarity with respect to the training data, they do not exhibit any visually striking degradation in the reconstructed mass maps, decontamination quality, or in the values returned by the regression head. This suggests that increasing D_M does not necessarily manifest as obvious artefacts or implausible predictions in the individual outputs.

Comparing the predicted masses and ages with the three independent literature catalogues (Fan et al. 2010; Chen et al. 2016; Usher et al. 2024) highlights both an overall consistency and an expected scatter.

- (a) For B386–G322, which is available in all three catalogues, our mass estimate ($\log M/M_{\odot} = 5.31$) lies between the values reported by Fan et al. (2010, $\log M/M_{\odot} = 5.28$), Chen

et al. (2016, $\log M/M_{\odot} = 6.47$), and Usher et al. (2024, $\log M/M_{\odot} = 5.89$). The corresponding age estimates show clear disagreement between the studies; π -DOC’s value of 9.43 Gyr agrees only with Usher et al. (2024) (8.26 Gyr), versus 1.28 Gyr from Fan et al. (2010) and 17.39 Gyr from Chen et al. (2016).

- (b) For B186, which is only present in Fan et al. (2010) and Chen et al. (2016), our mass ($\log M/M_{\odot} = 4.78$) is lower than both literature values ($\log M/M_{\odot} = 5.40$ and 5.39, respectively), while the age (12.64 Gyr) is similar to Fan et al. (2010, 12.75 Gyr) and older than Chen et al. (2016, 5.88 Gyr).
- (c) For B206, our mass ($\log M/M_{\odot} = 5.43$) is again lower than the literature estimates ($\log M/M_{\odot} = 5.62$, 6.21, and 6.10 in Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024), respectively), whereas for the age we again only find an agreement with Usher et al. (2024) (9.58 Gyr vs 1.61 Gyr, 5.01 Gyr, and 9.74 Gyr).
- (d) The fourth object, B371–G303, has no counterpart in these three catalogues, so no direct comparison is possible.

Expressed as linear mass ratios, the literature-to- π -DOC factors for the three clusters span roughly a factor of 1.5 to 15, with the largest discrepancy for B386–G322 against Chen et al. (2016) (factor ~ 14.6) and more moderate differences for the other cases (factors ~ 1.0 –6.0). Alongside the analysis of the performance on the synthetic data (Section 4), it seems likely that π -DOC is underestimating these masses, and particularly for B386–G322 and B206–G257.

The inferred distances span more than 100 kpc, from ~ 718 kpc (B206–G257) to ~ 833 kpc (B186), and appear to correlate with cluster brightness: the brighter B206–G257 and B386–G322 lie at smaller distances, while the fainter B186 and B371–G303 are placed further away. However, this trend is not conclusive. If the more distant clusters were simply on the far side of M31, one might expect higher extinction, yet Fan et al. (2010) find similar $E(B - V)$ for B186 and B386 (0.24 and 0.21 mag).

The clusters show a small morphological diversity. B386–G322 is spherical, while the others have ellipticities of $e \sim 0.07$ –0.08. For B206–G257, we find close agreement with Staneva et al. (1996, 0.085 vs. 0.086), whereas for B386–G322 we do not (~ 0 vs. 0.10). For B206–G257, the position angle differs by roughly 40° (67° vs. 24°), which is significant; however,

²⁰ With MC dropout enabled, the corresponding speed is ~ 0.7 s per GC for π -DOC₃ and ~ 0.5 s per GC for π -DOC₂

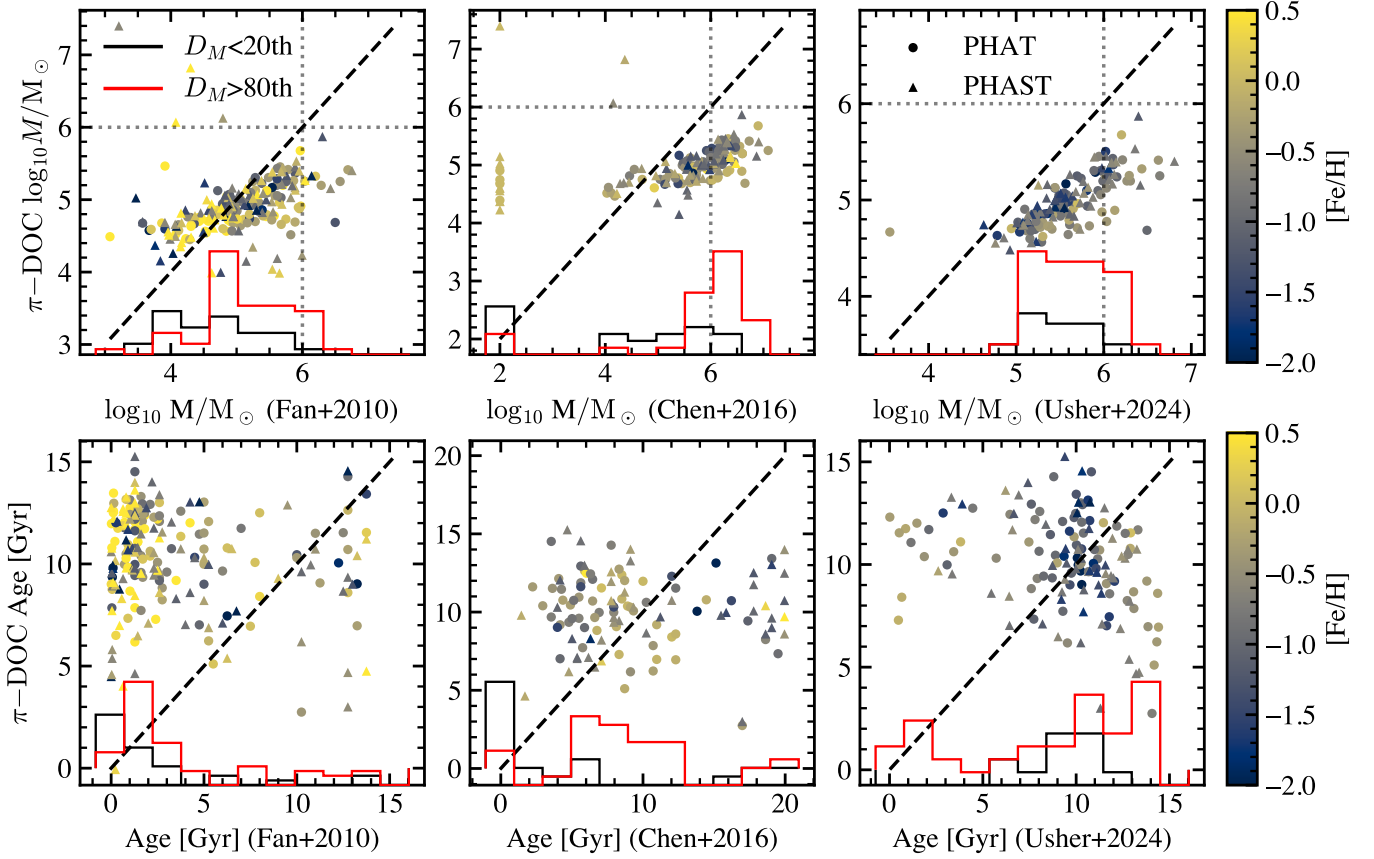


Fig. 10. Comparison of the π -DOC mass and age estimates with those of Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024), for 206, 143, and 141, GCs, respectively. The dashed black line indicates a 1:1 relation, and the dotted lines mark $10^6 M_\odot$, above which reliable coverage is not expected (see Figure 1). Points are colour-coded by the metallicity estimates from respective study. We find overall agreement in total mass to within a factor of about four, while acceptable age agreement is obtained only for low-metallicity GCs with Usher et al. (2024). The black histogram corresponds to GCs within the 20th quantile and the red histogram corresponds to GCs outside the 80th quantile of D_M .

this discrepancy is of limited concern given the cluster’s low ellipticity, for which the position angle is only weakly constrained.

6.2. Inference of GC properties within the PHAT and PHAST footprints

Figure 9 shows the GCs within the PHAT and PHAST footprints, colour-coded by the mass (a), age (b), ellipticity (c), and distance (d), as estimated by the respective π -DOC model. Overall, we find that the mass distribution is relatively uniform, although more massive GCs tend to lie closer to the M31 centre. We find a mean mass gradient of $-1186^{+244}_{-510} M_\odot \text{ arcmin}^{-1}$ with increasing projected distance from the centre. The uncertainties are estimated from 50k bootstrap realisations of the GC sample, with the 16th and 84th percentiles of the resulting slope distribution adopted as the lower and upper uncertainties, respectively. This trend could be consistent with orbital decay driven by dynamical friction, which preferentially affects massive clusters in the inner regions, but may also reflect selection effects or systematic uncertainties in the π -DOC mass estimates and the biases discussed above.

Most of the GCs are old ($\gtrsim 10$ Gyr) and appear relatively circular, with only a few exceeding an ellipticity of 0.1; these are predominantly located well outside the central regions of M31. This might possibly reflect a bias in underestimating ellipticity

measurements for GCs in a background-dominated regime (see Section 4.3), i.e. closer to the central regions.

We find an average distance 813 ± 31 kpc to the GCs in M31, with a slight difference between PHAT (805 ± 27 kpc)²¹ and PHAST (825 ± 33 kpc). A direct comparison of the networks’ estimates for PHAT GCs is also included in Appendix B, which confirms overall agreement while showing that π -DOC₂ typically infers greater distances than π -DOC₃ for the same input (see Figure B.1).

6.3. Mass and age literature comparisons

We proceed to compare the full sample of GCs to the three independent studies in Figure 10. For each respective study, we have an overlapping sample of 206 (Fan et al. 2010), 143 (Chen et al. 2016), and 141 (Usher et al. 2024) GCs, respectively.

We find that the comparisons in Section 6.1 were fairly representative of the overall agreement with these independent studies. For the mass, we find a clear rank-order agreement: clusters identified as more massive in the literature are also assigned higher masses by π -DOC, and vice versa. However, there is a systematic offset in the absolute scale, with π -DOC’s masses increasingly lower than literature values at the high-mass end ($\gtrsim 10^6 M_\odot$). This is particularly clear for individual cluster com-

²¹ When employing π -DOC₂ on the PHAT survey, we find (820 ± 32 kpc).

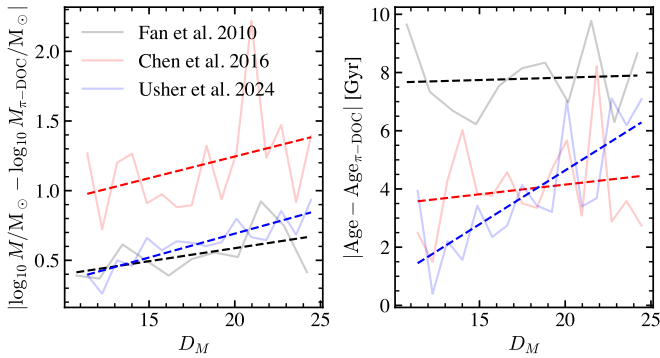


Fig. 11. Absolute differences in mass estimates (left panel) and age estimates (right panel) between π -DOC and Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024), as a function of the Mahalanobis distance D_M .

comparisons with Chen et al. (2016) and Usher et al. (2024). It is, furthermore, reflected in the average GC mass ($\log_{10}(M/M_{\odot}) = 4.72 \pm 0.5$ for our full sample and $\log_{10}(M/M_{\odot}) = 4.96 \pm 0.36$ for overlapping sample) which is a factor of 1.3–4 $\times$ lower than the corresponding mean mass derived from the overlapping samples with Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024). This is also in line with the biases seen for the synthetic dataset tests in Sections 4 and 4.3 (see also Figure 5), where the most massive clusters are underestimated up to a factor of around four.

For the ages, we also find a fairly good agreement with the overall distribution of Usher et al. (2024), whereas we find a very poor agreement with Fan et al. (2010) and Chen et al. (2016). We note that there is not a clear agreement among the three independent studies either.

6.3.1. D_M as a proxy for prediction reliability

In order to address the reliability of our estimates for real GCs, we conduct a similar exploration of the correlation with D_M to that presented in Section 5. We consider the differences in mass and age estimates between π -DOC and Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024).

Each panel in Figure 10 further indicates the distribution of GCs below the 20th percentile and above the 80th percentile of D_M , representing the most similar and least similar GCs to the training data, respectively. The least similar GCs are shifted towards higher masses, in line with the poor training-set coverage of very massive GCs discussed above. For Usher et al. (2024) there is, in addition, a much more narrow distribution for the age (roughly centred at the region where we find the most agreement). The most pronounced shift, however, occurs for metallicity ([Fe/H]): for Fan et al. (2010), the below-20th-percentile and above-80th-percentile groups have means and standard deviations of -0.64 ± 0.89 and -0.20 ± 0.73 dex, respectively; for Chen et al. (2016), -0.58 ± 0.57 and -0.31 ± 0.32 dex; and for Usher et al. (2024), -1.20 ± 0.48 and -0.72 ± 0.43 dex. In other words, the least similar GCs are systematically more metal-rich. This likely also explains our comparatively better agreement for age with Usher et al. (2024), whose inferred metallicities are closer to those of our training data ([Fe/H] ~ -1.3 ; Askar et al. 2025; Zhao et al. 2026; Bianchini et al. 2026; Marklund et al. 2026, see Section 2.2).

Figure 11 further supports the discussion above. We find that agreement with literature values improves continuously with de-

creasing D_M across the full sample, for all three comparisons. The corresponding Pearson correlation coefficients between D_M and the discrepancy with π -DOC are $r_p = 0.49$ (mass) and 0.06 (age) for Fan et al. (2010); 0.36 (mass) and 0.17 (age) for Chen et al. (2016); and 0.85 (mass) and 0.76 (age) for Usher et al. (2024). This reinforces D_M as a proxy for estimate reliability on real data, though we note one notable difference from the synthetic case: age discrepancy correlates strongly with D_M for Usher et al. (2024) (and, more weakly, if at all, for the other two studies), whereas no such relationship was found for age on the synthetic test set.

Since the real and synthetic comparisons differ both in the underlying data distribution and in what is being measured (discrepancy with an independent literature estimate, rather than error against known ground truth), we do not consider this a contradiction; it may instead reflect physical correlations that are not represented in our synthetic training grid. We leave a more detailed investigation of this discrepancy to future work.

6.3.2. Methodological discrepancies

Part of the discrepancies with the literature studies may also arise from differences in methodology and observational data. Usher et al. (2024) combined ground-based photometry from the Sloan Digital Sky Survey (SDSS; York et al. 2000) and the Pan-Andromeda Archaeological Survey (PAndAS; Huxor et al. 2014), and with spectroscopy from Caldwell et al. (2009); Sakari & Wallerstein (2016, 2022); Sakari et al. (2021) where they target the CaT region.

Using Markov Chain Monte Carlo (MCMC), they simultaneously fitted ages, metallicities, present-day stellar masses, and reddening with FSPS models (Conroy et al. 2009; Conroy & Gunn 2010) based on MIST isochrones (Choi et al. 2016). Explicit reddening modelling systematically increases the inferred total brightness and thus the associated mass estimates compared to our values, since we only account for Milky Way reddening. In addition, their photometric masses are derived from stellar-population models that do not explicitly account for the dynamical evolution of the cluster. By contrast, π -DOC is trained on dynamically evolved clusters and can therefore encode changes to the present-day stellar mass function and remnant population resulting from dynamical processes such as mass segregation, evaporation, and tidal stripping. Differences in these underlying assumptions may therefore contribute to the discrepancies between the two mass estimates, consistent with the known differences between photometric and dynamical mass estimates (Sollima et al. 2017).

Using Markov Chain Monte Carlo (MCMC), they simultaneously fitted ages, metallicities, present-day stellar masses, and reddening with FSPS models (Conroy et al. 2009; Conroy & Gunn 2010) based on MIST isochrones (Choi et al. 2016). Explicit reddening modelling systematically increases the inferred total brightness and thus the associated mass estimates compared to our values, since we only account for Milky Way reddening. At the same time, their photometric masses trace the current surviving stellar populations and do not include dynamical mass loss (e.g. stellar-mass dependent evaporation and stripping), which π -DOC does; photometric and dynamical mass estimates are known to show discrepancies (Sollima et al. 2017).

Chen et al. (2016) also combined ground-based photometry from Peacock et al. (2010a) with spectroscopy using the Large Sky Area Multi-Object Fibre Spectroscopic Telescope (LAMOST Liu et al. 2015; Yuan et al. 2015). Ages and metallicities were derived via pixel-by-pixel fitting of integrated spec-

tra to single stellar population (SSP) models, with masses inferred from multi-band photometric spectral energy distributions (SEDs) scaled to those parameters. Similarly, Fan et al. (2010) also derived age and mass estimates by comparing their observed SEDs with SSP models. Their estimates thus yield photometric masses and are subject to similar modelling assumptions as those in Usher et al. (2024).

We have already identified the metallicity distribution as a likely contributor to the poor overall agreement in age. However, studies that agree more closely in metallicity, such as Chen et al. (2016) and Usher et al. (2024), exhibit scatter comparable to that seen with π -DOC, highlighting the substantial discrepancies between independent literature estimates (see also Usher et al. 2024). The presence of ages near 0 Gyr or 20 Gyr in Fan et al. (2010) and Chen et al. (2016), respectively, further illustrates the challenges associated with determining GC ages. These comparisons should also be interpreted in light of the systematic uncertainties associated with the underlying stellar-population models. In particular, ages derived from SSP fitting are subject to degeneracies between age, metallicity, and extinction, as well as differences between SSP model prescriptions (Worthey 1994, 1999; Usher et al. 2019, 2024). Conversely, π -DOC inherits assumptions from the simulations used for training, including the adopted stellar-evolution models and the limited range of metallicities represented in the training set. Such differences in the underlying modelling assumptions may contribute to some of the discrepancies between π -DOC and the literature values.

These comparisons therefore suggest that accurate age estimation remains challenging across different methodologies, rather than pointing to specific shortcomings of π -DOC or any individual literature study. Importantly, the correlation between metallicity and D_M identified from comparisons of the 20th and 80th percentiles of the D_M distribution suggests that metallicity information may already be encoded in the network’s learned features. This raises the possibility of extending future versions of π -DOC to predict age and metallicity jointly as explicit outputs, potentially providing a new approach to addressing the age–metallicity degeneracy.

6.4. Literature comparisons of GCs’ morphologies

We compare our ellipticity and position angle estimates with Staneva et al. (1996) in Figure 12, where we have a total overlapping sample of 103 GCs. There is a broad agreement in the overall ellipticity distribution, but π -DOC generally infers lower values. Discrepancies are to be expected given the differences in methodology: Staneva et al. (1996) derived ellipticities from iso-density contours of ground based photometry, whereas π -DOC was trained on the intrinsic stellar distributions within the field of view, quantified using the second-moment tensor method (see e.g. Marklund et al. 2026, for details). These two approaches are known to produce systematically different results even for identical stellar distributions (Fréour et al. 2026; Marklund et al. 2026). Furthermore, Fréour et al. (2026) demonstrated that ellipticity measurements for more spherical distributions ($e \lesssim 0.1$) are known to produce large systematic overestimates and that these biases need to be corrected, which is not done for (Staneva et al. 1996).

A second important factor is spatial resolution and coverage. Ground-based observations inevitably have lower resolution and typically a wider FoV than HST imaging. Outer cluster regions, which dominate ground-based measurements, are generally more elliptical than inner regions (Fréour et al. 2026; Marklund et al. 2026). For spatially resolved clusters in the Milky

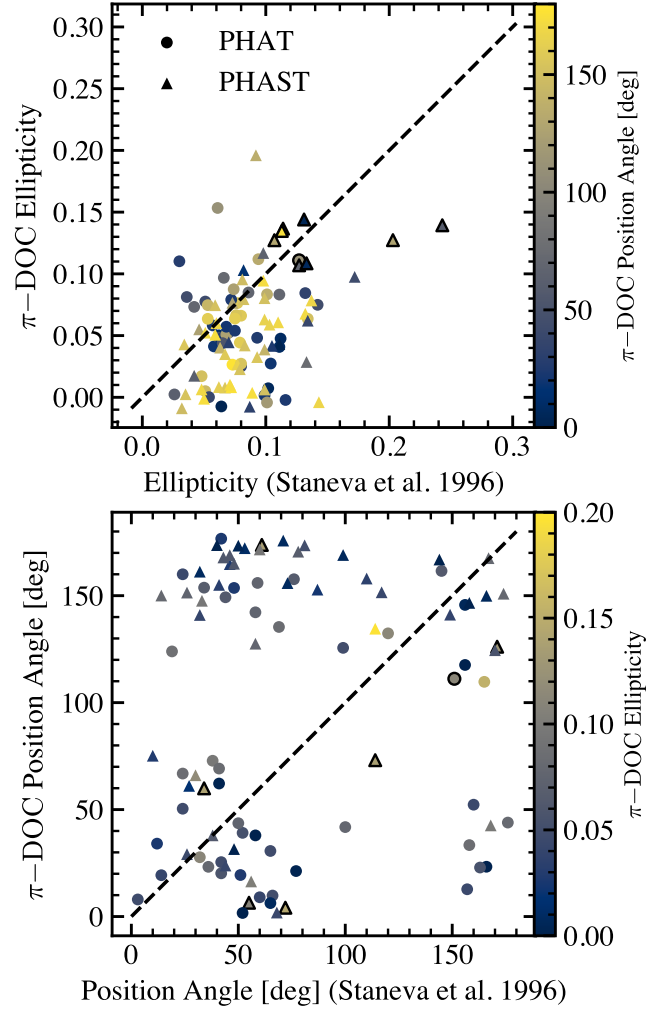


Fig. 12. Comparison between π -DOC’s ellipticity and position angles estimates for 103 GCs to those in Staneva et al. (1996). π -DOC’s ellipticity estimates are systematically lower than those of (Staneva et al. 1996), and the position angles show a poor overall agreement. This can partially be explained by the degeneracy at 0° , 180° and overall small values for ellipticity. The dashed black line indicates a 1:1 relation.

Way, Fréour et al. (2026) found that typically there is not a significant difference between the measured flattening within $3r_{50\%}$ and $19r_{50\%}$. On the contrary, the intrinsic flattening within and outside $r_{50\%}$ may show significant differences (Marklund et al. 2026). Although our mock images extend beyond the half-mass radius of the clusters, they remain confined to radii of approximately 20 – 30pc (set by the chosen FoV of our images and subject to the distance). This difference in spatial resolution and radial coverage may naturally contribute to the observed offsets.

Although direct one-to-one agreement in ellipticity across different studies should not be expected (Fréour et al. 2026), one might at least anticipate reasonable agreement in position angle for clusters identified as elliptical ($e \geq 0.1$) by both Staneva et al. (1996) and π -DOC. These points are highlighted with black edges in Figure 12, and show that differences exceeding 20–30° persist even in this regime. We attribute this to the position angle being inherently poorly constrained and sensitive to measurement choices — such as isophote level, background subtraction, and fitting method — any of which can shift the inferred orientation even when the underlying structure is similar (Fréour et al. 2026; Marklund et al. 2026).

7. Improving performance on massive GCs

We have suggested throughout this work that π -DOC is less reliable for the most massive GCs, primarily due to their limited representation in the training data — particularly for the MOCCA models. If this is correct, a network exposed to more such clusters during training should show improved performance in the high-mass regime ($\gtrsim 10^6 M_\odot$). To test this, we continue training π -DOC₃ for 10 additional epochs (~ 300 GPU hours) with two changes: (i) all loss weights (Section 3.4) are set to zero except decontamination ($w_{\text{decontamination}} = 1$) and mass ($w_{\text{mass}} = 10$), so the network focuses solely on these two outputs (and should not be employed for other quantities), contributing roughly equally to the total loss (Table 1); and (ii) each batch is constrained to include at least 25% MOCCA models, increasing their relative weight in training. We use a low learning rate (10^{-7}), as the network is fine-tuned from a previously trained model and therefore only requires small parameter updates for stable convergence. We denote the resulting network π -DOC₃^{MOCCA}, and stress that this is a proof-of-concept test rather than an optimised configuration: adjusting these hyperparameters further is likely to improve both performance and convergence speed.

Figure 13 shows that this fine-tuning shifts the predicted MOCCA masses upward (see panel a) and reduces the underestimation bias (see panel b) across all models, including the N -body ones. This latter improvement likely reflects the network training exclusively on decontamination and mass, rather than the increased MOCCA fraction specifically. For the synthetic test set, the recovered mean and 1σ of the mass distribution is $5.28^{+0.06}_{-0.08}$ dex for π -DOC₃ and $5.45^{+0.07}_{-0.08}$ dex for π -DOC₃^{MOCCA} (ground truth: $5.77^{+0.03}_{-0.05}$ dex). As a result, the systematic underestimation in the MOCCA models is reduced from a factor of 3 to a factor of 2. A similar shift is reflected in the PHAT sample (see Section 6 and in particular Section 6.3): $4.70^{+0.06}_{-0.13}$ dex for π -DOC₃ versus $4.82^{+0.05}_{-0.16}$ dex for π -DOC₃^{MOCCA}.

We also argued that a possible, and not mutually exclusive, explanation for these discrepancies is that the MOCCA models have a more centrally concentrated population of stellar remnants, resulting in a fundamentally different radial dependence of the mass-to-light ratio. This may be related to their shorter relaxation times, which allow more efficient mass segregation and consequently lead to a stronger central concentration of stellar remnants. Figure 14 shows that this is the case: (i) both the N -body and MOCCA models are dominated by remnants in their central regions, but the ratio between the remnant and visible mass components is markedly more extreme for the MOCCA model; (ii) π -DOC's predictions closely track the visible mass rather than the total mass, so the underestimated mass is concentrated precisely in the remnant-dominated regions; (iii) the improved mass estimate on the MOCCA model using π -DOC₃^{MOCCA}, although marginal, is applied roughly uniformly across radius rather than being concentrated in the remnant-dominated core — suggesting the fine-tuning corrects the network's overall visible-to-total mass scaling rather than specifically learning to identify the remnant-dominated regions themselves. A promising extension would be to explicitly condition the pixel-wise mass correction on global cluster properties (e.g. via attention or feature-wise modulation), allowing the network to apply a context-dependent M/L correction.

These results confirm that increasing the relative fraction of MOCCA models in training improves performance in the high-mass regime, and suggest more generally that π -DOC can be fine-tuned toward specific tasks — for example, improving the distance estimates, which we know to be biased toward the mean

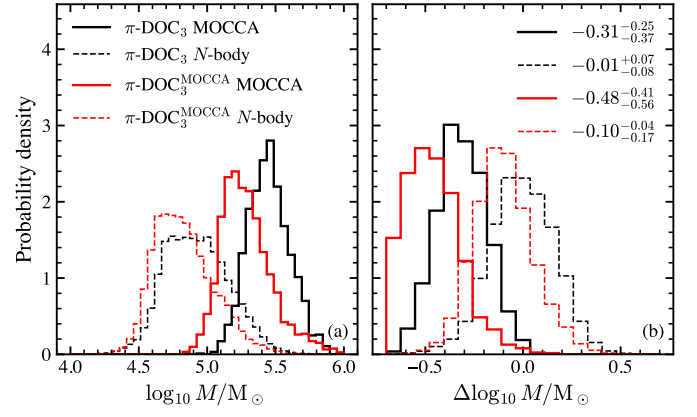


Fig. 13. Recovered probability density distribution for the mass (panel a) and corresponding errors (panel b) for MOCCA (solid lines) and N -body (dashed lines) mock images using π -DOC₃ (black lines) and π -DOC₃^{MOCCA} (red lines). The labels in panel (b) indicate the mean and $\pm 1\sigma$ of the corresponding error distribution.

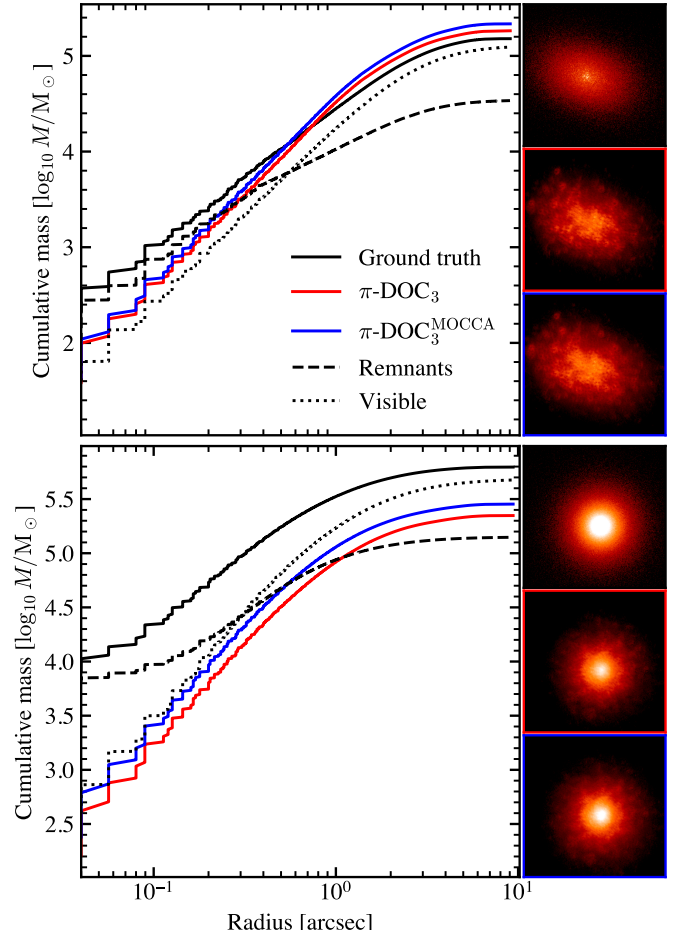


Fig. 14. Cumulative mass profiles. Top row: N -body mock cluster (a) in Figure 4. Bottom row: MOCCA mock cluster (b) in Figure 4. Solid black line shows the ground truth while the solid red and blue lines show the predicted mass maps from π -DOC₃ and π -DOC₃^{MOCCA}, respectively. Corresponding profiles for only remnants (dashed black line) and only visible mass (dotted black line) are also indicated, with 2D mass maps shown on the right, ordered from top to bottom: ground truth, predicted by π -DOC₃, predicted by π -DOC₃^{MOCCA}, respectively.

of the training distribution (see Figure 5). For this paper, we prioritised a general-purpose model with balanced performance across all outputs, but future versions could instead be fine-tuned for specific tasks. One promising application would be the identification of black hole signatures: several of our MOCCA models host intermediate-mass black holes, and all of our simulations contain stellar-mass black holes. A dedicated network could therefore, in principle, be trained to identify their observational signatures.

8. Conclusions

In this work, we introduced two complementary frameworks: CREMANT, a forward-modelling framework that transforms simulation snapshots into realistic mock observations (see Section 2), and π -DOC, a neural network that infers intrinsic dynamical and morphological GC properties from multi-band photometric images and pixel CMDs (see Section 3). Both were developed aiming at an application to HST observations of M31 from the PHAT and PHAST surveys (Dalcanton et al. 2012; Chen et al. 2025). π -DOC performs field-star decontamination, reconstructs the 2D mass distribution, and infers age, distance, ellipticity, and position angle.

The algorithm is trained on some of the most realistic GC simulations (ROLLIN' Bianchini et al. 2026; Marklund et al. 2026 and MOCCA Askar et al. 2025; Zhao et al. 2026), guaranteeing, together with the CREMANT forward modelling framework, a direct applicability to real observations. The inference speed of π -DOC is ~ 10 GCs/s, processing the GCs within the combined PHAT and PHAST footprint in well under a minute — illustrating the scalability of this forward-modelling and deep learning approach for future surveys.

In synthetic tests representing a best-case scenario (see Section 4), π -DOC performs well. With the 2σ limit as a conservative confidence, we found that ages are recovered to within $\lesssim 3.6$ Gyr, with masses typically constrained to within a factor of 2-3. A systematic mass underestimation persists for the most massive clusters ($\gtrsim 10^6 M_\odot$), likely due to their limited training-set representation. Ellipticity and position angle are fairly well constrained for flattened clusters, with a more significant scatter at low ellipticity ($\lesssim 0.1$); distance estimates remain comparatively uncertain ($\gtrsim 80$ kpc). Decontamination is most reliable when the GC is brighter than the background ($S/B < 0$), but remains fairly robust even in background-dominated regimes ($S/B > 0$), indicating π -DOC could be re-purposed for GC detection. The inferred properties are generally recovered to within ~ 0.85 mag, depending on the passband, although a systematic bias of up to ~ 0.6 mag is present in some cases, resulting in a non-negligible bias in the inferred colours.

Applied to the PHAT and PHAST samples (see Section 6), π -DOC yields mass estimates broadly consistent with the literature, though systematically lower by a factor of 1.3-4 relative to Fan et al. (2010), Chen et al. (2016), and Usher et al. (2024) — consistent with the high-mass underestimation seen in the synthetic tests. Interestingly, our results also suggests an average decrease in the GCs' mass of $1186^{+244}_{-510} M_\odot \text{ arcmin}^{-1}$ per arcminute from the centre of M31 (see Section 6.2). This could be interpreted as a consequence of dynamical friction, but may also reflect the intrinsic biases and uncertainties of π -DOC.

The average distance estimate, 813 ± 31 kpc, is somewhat larger than typical M31 values (e.g. McConnachie et al. 2005), though consistent within uncertainties, while age estimates show the expected scatter from the age-metallicity degeneracy and the training set's metal-poor composition, with better agreement at

low metallicity. The ellipticity distribution agrees with Staneva et al. (1996) in aggregate, though not always cluster-by-cluster, likely reflecting known measurement biases between space- and ground-based imaging (Fréour et al. 2026). Together, these comparisons provide external validation of our approach.

We compared the latent-space representation of the observations (for mock test data, see Section 5, for real data, see Section 6) to the training data in order to quantify the reliability of π -DOC's estimates via the Mahalanobis distance (D_M). We found that D_M typically correlates with agreement to independent studies and estimate reliability: the least similar clusters are systematically more massive, and especially more metal-rich, than the most similar ones — the latter plausibly linked to D_M 's relationship with cluster colour. This reflects the limited training-set coverage for massive and metal-rich clusters, and identifies the model's main current limitation, motivating more extensive simulation grids exploring this parameter space. It also suggests that colour and metallicity information already present in the network's latent space could, with dedicated training, be exploited to jointly constrain age and metallicity in future extensions.

Preliminary tests also suggest π -DOC can be optimised for specific inference tasks (e.g. high-mass robustness, which plausibly could be extended to stellar remnant fraction detection, see Section 7), potentially reducing the need for very large training sets. Future developments will focus on extending the applicability of π -DOC beyond HST data of semi-resolved GCs. We expect retraining to generalise well to other setups, though adapting across telescopes and host galaxies remains an open challenge; domain-adaptation techniques (Thuruthipilly et al. 2026) suggest very large survey-specific datasets may not be necessary. Future work should explore such techniques and a more flexible architecture (e.g. Parker et al. 2024, 2025), robust to passband combinations (e.g. via random band dropout) and image size, as required for more distant globular clusters.

Our work demonstrates that a forward modelling and deep learning framework offers a scalable and physically grounded approach to estimating GC properties from large photometric surveys. The discrepancies we find with independent studies largely reflect challenges shared across all methods — most notably in age determination — rather than shortcomings unique to π -DOC, while the coverage gaps in our training data, particularly at high mass and metallicity, indicate the clearest path for future improvement. We expect that our framework and others alike will play a growing role in the analysis of GC populations in the coming era of large-scale imaging surveys.

Data availability

The data and the GitHub repositories will be made publicly available upon acceptance of the manuscript.

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Appendix A: Realism of mock images

Figure A.1 shows a qualitative comparison between five mock images (top row) and five GCs from the PHAT survey (middle row). The bottom row shows the corresponding pixel-CMDs. The clusters resemble each other in appearance: similar sizes, shapes, and brightness. Mock images were chosen based on their visual resemblance, and may not correspond to physical similarity. There are notable differences between the images, especially noticeable in the pixel-CMDs, for example with respect to the base level (governed by the background) and colour (mainly governed by the age and metallicity). Each RGB image is scaled identically to the inputs in Figure 4.

Appendix B: π -DOC in two versus three passbands

We use two versions of π -DOC for the survey analyses: π -DOC₃ for PHAT and π -DOC₂ for PHAST, the latter being limited to two passbands. To assess the effect of the missing F336W passband, we directly compare the two networks and their predictions for the mock images in our test set. Figure B.1 shows that the networks generally agree within the typical confidence limits (Section 4.3).

A similar comparison for the PHAT GCs reveals the same overall trends. In general, the two networks agree reasonably well for each quantity, although with a non-negligible scatter. The mean absolute discrepancies (differences computed as π -DOC₂ minus π -DOC₃) are approximately 0.28 dex for the luminosity in both passbands (F475W and F814W), 0.24 dex for mass, 1.3 Gyr for age, 23.5 kpc for distance, 0.02 for ellipticity, and 66° for position angle. Agreement is particularly good for age across the full range of values. For luminosity, however, π -DOC₃ tends to predict brighter values for the faintest objects, suggesting that the third passband is especially informative for low-luminosity GCs; note that this regime is close to the edge of the coverage by the mock test data. At the bright and massive ends, the opposite trend is seen, with π -DOC₂ typically predicting luminosities and masses a factor ~ 1.5 higher than π -DOC₃. Distances agree well overall, although π -DOC₃ more often predicts shorter distances, possibly because the additional pixel CMDs improve the distance inference. Ellipticity is concentrated near $e \sim 0.1$, indicating that it is only weakly constrained; this is consistent with the synthetic tests, where nearly spherical clusters tend to be overestimated and objects with $e < 0.1$ show substantial scatter. Position angle shows the largest discrepancies, likely because most clusters in the sample are only mildly elliptical. By contrast, the synthetic tests in Section 4 showed that more elliptical clusters ($e \gtrsim 0.1$) are much better constrained, implying that such objects would likely have produced a clearer signal in the real-sample predictions.

Appendix C: Robustness to noise

Most of our clusters are faint, whereas most of the backgrounds are, on average, a few magnitudes brighter. The accuracy of the predictions should therefore depend on the relative brightness of the GC and the background. In Figure C.1, we show the two-dimensional distributions of the errors for each quantity as a function of the magnitude difference between the GC and the corresponding background in the F814W passband²². Negative values indicate that the GC is brighter, while positive values indicate that the background is brighter. Qualitatively, we find that

the errors depend on the relative GC-to-background brightness in much the same way as they do on GC brightness alone (see Figure 6): fainter clusters generally correspond to lower signal-to-background, and the spread in the predictions increases substantially once the background dominates.

The top row shows the magnitudes and total mass, both derived from their respective maps. At low signal-to-background, there is a significant bias in each quantity toward underestimating brightness and total mass. Even at high signal-to-background, the GC brightness remains systematically underestimated in F336W and F475W, with the smallest bias occurring near the most common GC-to-background brightness difference in the training data; the bias increases again toward both the lowest and highest signal-to-background. For F814W the bias is instead smallest at high signal-to-background and largest at low signal-to-background. The origin of this passband-dependent asymmetry is not clear from our analysis. The pattern is nevertheless suggestive of a colour-dependent bias in the decontamination, possibly linked to how the network blends GC and background differently across passbands at high relative brightness. Regardless, decontaminated clusters and their integrated magnitudes should be regarded as upper limits, since the true clusters are in most cases brighter than predicted by π -DOC.

The total mass shows no such systematic bias at high signal-to-background, but is instead associated with a larger absolute spread, of up to a factor of $\sim 3 - 4$. The regression-head outputs (bottom row) behave oppositely: they remain broadly symmetric around zero and are therefore unbiased. Only the ellipticity estimates show a bias to be underestimated in background-dominated images, where the stellar distribution becomes more diffuse and the cluster consequently appears more circular (the central regions of GCs are typically more circular; see Marklund et al. 2026). This effect is at most 0.03, small compared with the overall spread. Moreover, for GCs that are brighter than the background, the estimates are much better constrained, which is especially clear for age and ellipticity.

The low-signal-to-background mock images in Figure 4 already showed that the network is reasonably robust to noise (although π -DOC's estimates were less accurate than the corresponding high-signal-to-background examples). To explore this further, we investigate how the predictions change when the same mock image of a relatively faint cluster (F814W ~ 17.2) is combined with backgrounds of different magnitudes in Figure C.2, ranging from 0.8 to 3.8 mag brighter than the GC.

Most predictions remain relatively accurate and scatter around the ground truth, the magnitudes are in all cases except one overestimated, as expected from the discussion above (see Figure C.1). Overall, the predictions remain fairly robust to the varying noise levels. The total mass varies within a factor of ~ 1.8 , the position angle within $\sim 10^\circ$, the ellipticity within 0.09 (lower predicted values for more diffuse distributions where the GC inevitably appears more circular), the distance within ~ 50 kpc, the age within 6 Gyr (with the background-free and the faintest background lying within 0.8 Gyr to the ground truth), and the integrated magnitude within 0.6 mag in F814W. Except for the age, these variations lie within the confidence range previously defined for each quantity in Section 4.3, and also show that more reliable estimates can be expected for GCs in less crowded and less bright regions.

Appendix D: Epistemic Uncertainty

For each input image, we perform 20 stochastic forward passes (using an MC dropout rate of 0.1) and compute the mean and

²² We note that there are no substantial differences for the F336W and F475W passbands.

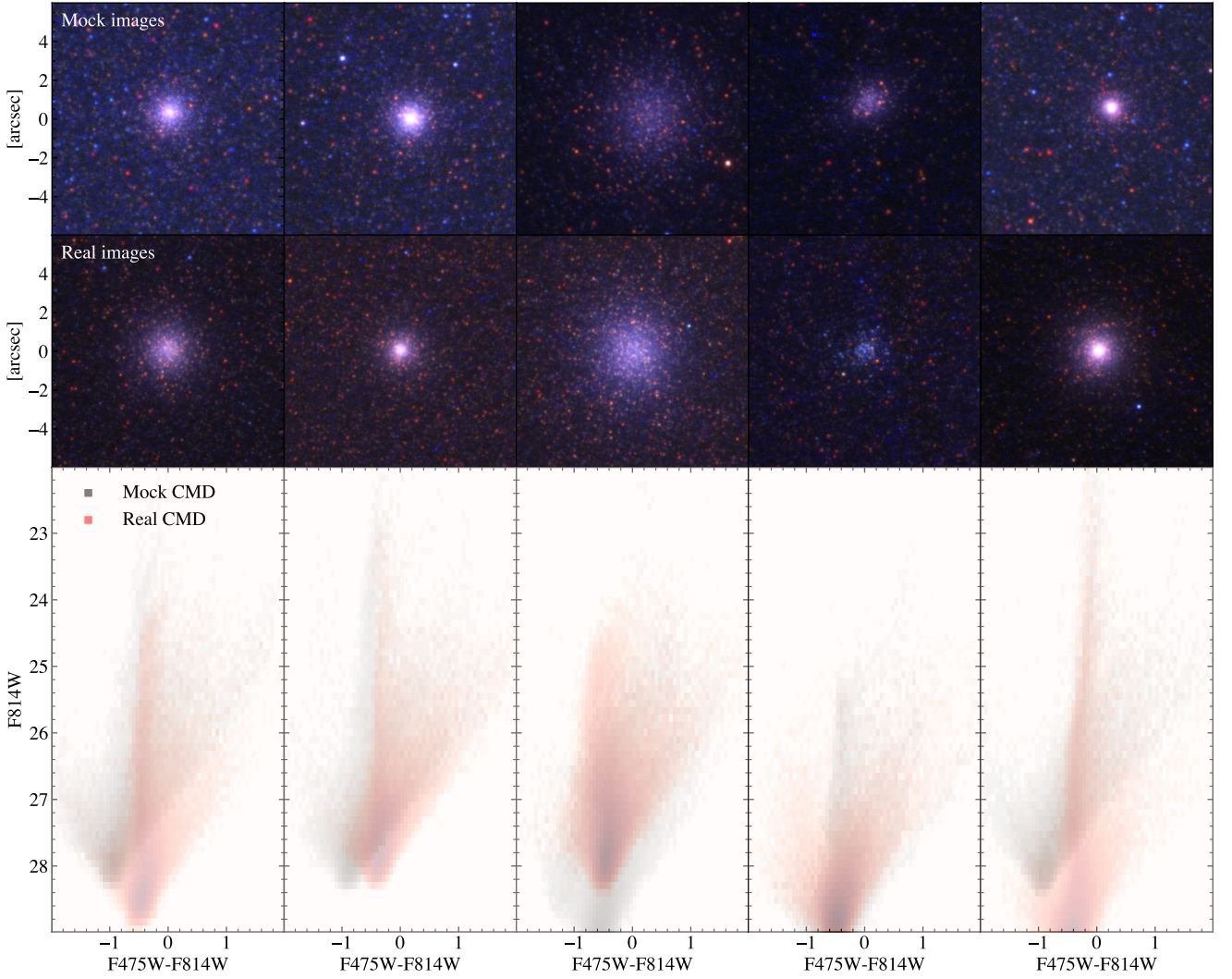


Fig. A.1. Visual comparison of mock images (top row), real images (middle row), and the corresponding pixel-CMDs (bottom row) for M31 GCs.

standard deviation of the resulting predictions. We adopt the mean prediction as the final model output, while the standard deviation is used to quantify the model’s confidence. This procedure naturally yields pixel-wise uncertainty maps for spatially resolved outputs (e.g. mass and magnitude maps), as well as uncertainties for our scalar regression targets: age, distance, ellipticity, and position angle. Because we predict the $\cos(2\text{P.A.})$ and $\sin(2\text{P.A.})$ components rather than the position angle directly, and to account for the circular nature of angular quantities, the epistemic uncertainty is estimated using the circular standard deviation. Specifically, we convert each sampled angle to its unit-vector representation ($\cos \text{P.A.}$, $\sin \text{P.A.}$), compute the mean resultant vector over stochastic forward passes, and derive the circular standard deviation as $\sigma_{\text{circ}} = \sqrt{-2 \ln R}$, where R is the magnitude of the mean resultant vector. The mean position angle is obtained from the averaged P.A. = $\frac{1}{2} \arctan 2(\sin(2\text{P.A.}), \cos(2\text{P.A.}))$.

The resulting epistemic uncertainties are shown in Figure D.1 against the absolute error of the predictions for the test set. We see that the epistemic uncertainties, for all predicted quantities, significantly underestimates the true error. As such, it does not tell us much about the physical uncertainty in the prediction of a cluster, but rather indicates the confidence level of π -DOC’s prediction.

Appendix E: Catalogue of properties for GCs in M31

π -DOC’s estimates for the various quantities discussed throughout this paper are indicated in Table E.1 for a subset of the GCs in the PHAT and PHAST footprints.

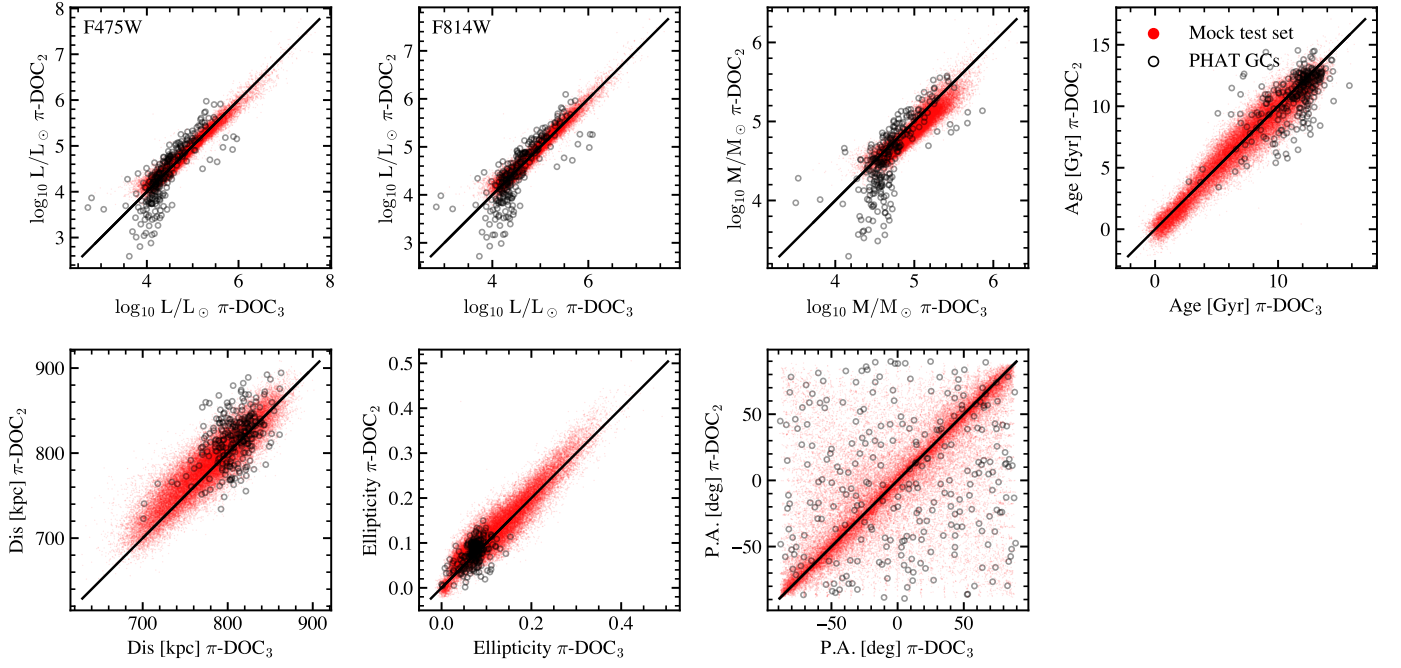


Fig. B.1. Comparisons of estimates between π -DOC₃ (x-axis) and π -DOC₂ (y-axis) for the total luminosities (within FoV) in F475W and F814W passbands, total mass (within FoV), age, distance, ellipticity, and position angle. Overall, the networks agree (black line indicates a 1:1 correspondence), however, they show some intrinsic scatter with respect to each other for the mock test set (red points). This scatter is also seen for the estimates of M31's GC properties which otherwise follow the distributions of the test set.

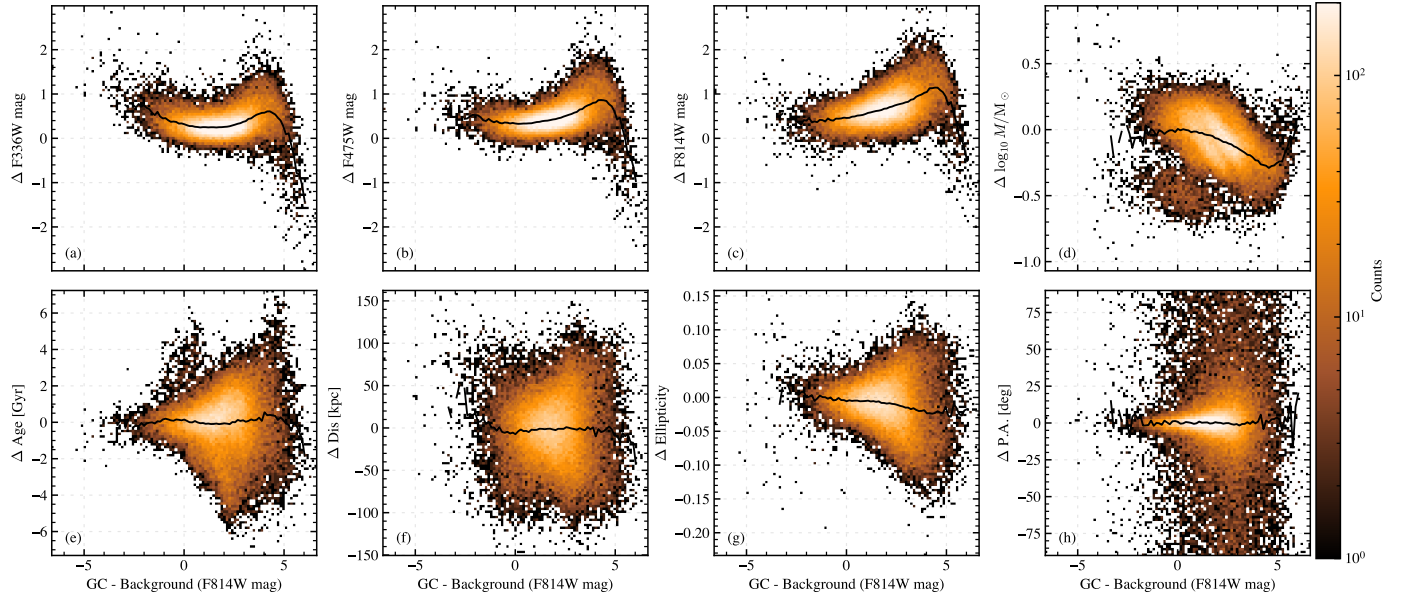


Fig. C.1. Difference between predicted value and ground truth for every quantity and how they depend on the relative brightness between a cluster and its associated background. π -DOC is less accurate for background dominated images, indicating that low signal-to-background, in particular, is a limitation. The solid black line indicates a moving mean.

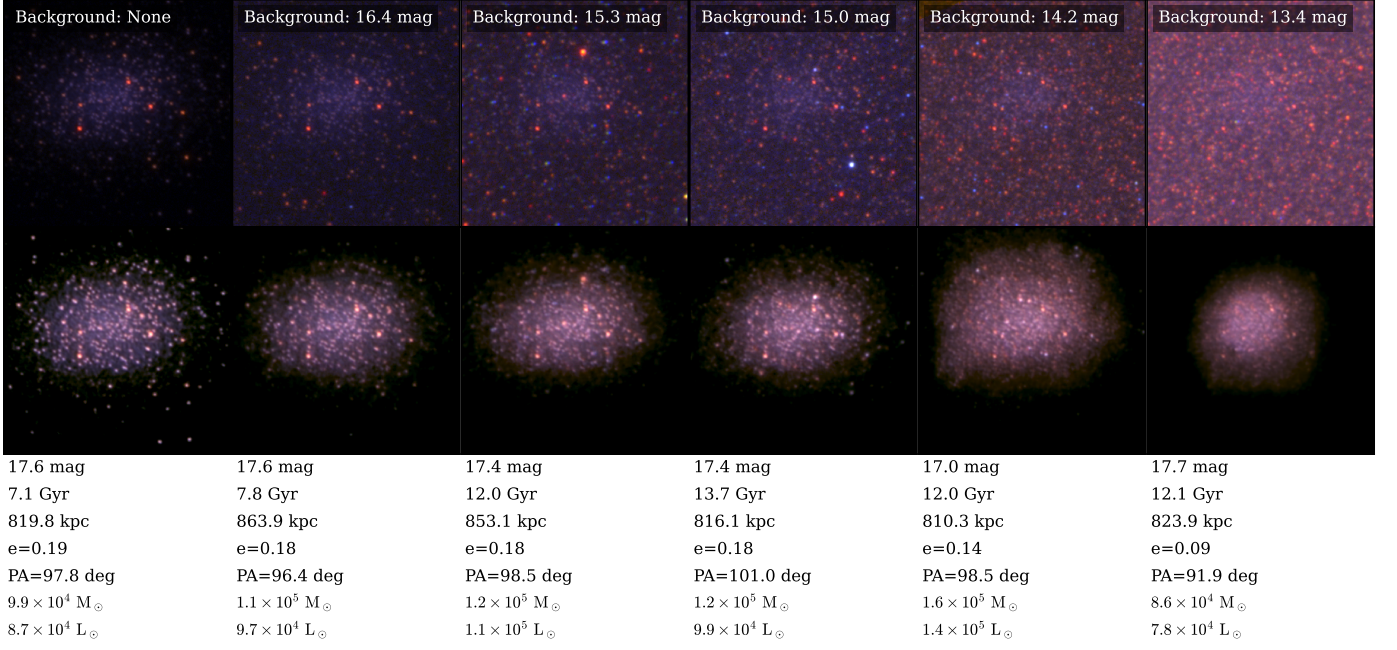


Fig. C.2. One mock GC combined with different backgrounds (top row) scaled identically as the inputs in Figure 4, and π -DOC's predictions (bottom row), scaled identically as the decontaminated images in Figure 4. Each mock image's associated predictions are indicated beneath the decontaminated image. The ground truth, in the same order as the predictions from top to bottom, are: $1.4 \times 10^5 M_{\odot}$, $1.3 \times 10^5 L_{\odot}$, $\phi = 97.9^{\circ}$, $e = 0.20$, 842.8 kpc, 8.0 Gyr, and the total integrated magnitude in the F814W passband is 17.2 mag. The predictions remain within the corresponding confidence limits, and the decontamination is fairly robust to decreasing signal-to-background.

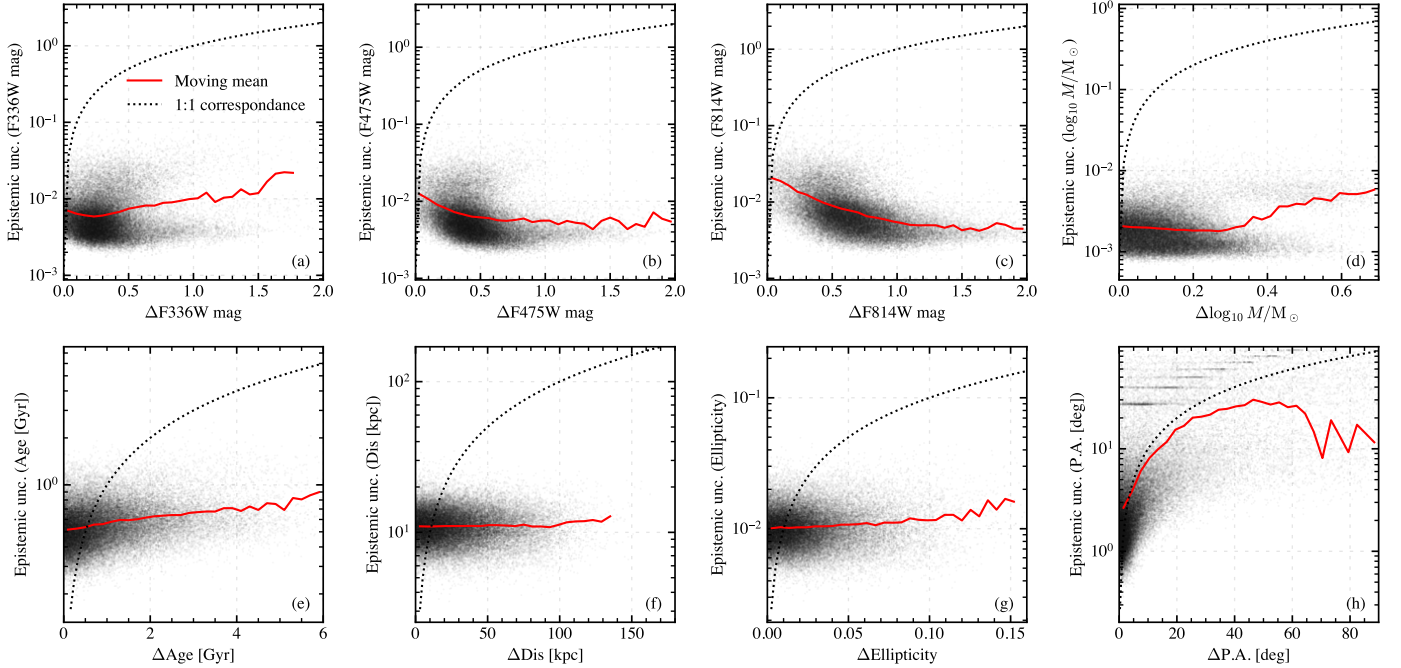


Fig. D.1. Epistemic uncertainty compared to actual errors for each predicted quantity. In all cases, the epistemic uncertainty significantly underestimates the error.

Table E.1. Globular cluster catalogue.

Name	Ra deg	Dec deg	F336W mag	F475W mag	F814W mag	log $M/M_{\odot}$ dex	Age Gyr	Distance kpc	Ellipticity	P.A. deg	D_M
SK122C	10.82	41.51	20.09 ± 0.00	19.07 ± 0.00	19.27 ± 0.01	4.54 ± 0.00	13.08 ± 0.55	815.84 ± 11.36	0.07 ± 0.01	131.47 ± 38.04	16.30
ACH 12	10.76	41.21	19.75 ± 0.02	18.68 ± 0.02	18.92 ± 0.01	4.61 ± 0.00	12.84 ± 0.55	802.16 ± 7.13	0.08 ± 0.01	41.98 ± 14.50	20.66
NB35-AU4	10.64	41.31	19.61 ± 0.01	18.61 ± 0.01	18.91 ± 0.01	4.60 ± 0.00	12.12 ± 0.38	799.15 ± 17.70	0.08 ± 0.01	47.21 ± 51.46	24.02
NB21-AU5	10.66	41.27	16.56 ± 0.00	15.49 ± 0.00	15.73 ± 0.00	5.46 ± 0.00	11.97 ± 0.65	808.55 ± 7.18	0.06 ± 0.01	66.99 ± 10.04	19.29
V129-BA4	11.44	41.87	18.41 ± 0.01	17.40 ± 0.01	17.75 ± 0.01	4.86 ± 0.00	9.81 ± 0.66	778.87 ± 10.40	0.04 ± 0.01	59.02 ± 5.85	16.71
B103-G165	10.62	41.30	17.18 ± 0.02	16.05 ± 0.02	16.16 ± 0.02	5.20 ± 0.00	7.86 ± 1.00	779.56 ± 9.35	0.05 ± 0.01	9.81 ± 6.93	22.72
B104-NB5	10.62	41.29	18.43 ± 0.02	17.38 ± 0.02	17.65 ± 0.03	4.83 ± 0.01	12.08 ± 0.99	808.71 ± 13.82	0.04 ± 0.02	15.64 ± 77.88	23.97
B107-G169	10.63	41.33	17.24 ± 0.01	16.13 ± 0.01	16.35 ± 0.01	5.10 ± 0.00	7.72 ± 0.82	801.63 ± 14.32	0.04 ± 0.01	19.32 ± 59.76	19.23
B112-G174	10.64	41.30	18.76 ± 0.03	17.71 ± 0.03	17.92 ± 0.03	4.78 ± 0.01	9.19 ± 0.89	790.40 ± 15.85	0.07 ± 0.02	53.23 ± 8.36	24.06
B115-G177	10.64	41.23	18.34 ± 0.02	17.18 ± 0.02	17.35 ± 0.02	4.91 ± 0.00	8.41 ± 0.99	821.37 ± 14.77	0.11 ± 0.01	178.59 ± 10.88	23.06
B119-NB14	10.65	41.29	18.71 ± 0.02	17.65 ± 0.02	17.90 ± 0.03	4.79 ± 0.00	12.38 ± 1.05	815.45 ± 15.68	0.06 ± 0.03	135.05 ± 109.79	23.88
B124-NB10	10.67	41.26	15.73 ± 0.01	14.61 ± 0.01	14.81 ± 0.01	5.68 ± 0.00	7.29 ± 0.61	813.92 ± 12.60	0.09 ± 0.01	64.34 ± 7.37	22.98
B126-G184	10.68	41.21	18.46 ± 0.01	17.45 ± 0.01	17.77 ± 0.01	4.79 ± 0.00	13.30 ± 1.63	832.34 ± 12.32	0.03 ± 0.02	164.14 ± 7.77	23.92
B129	10.70	41.42	19.77 ± 0.02	18.48 ± 0.02	18.44 ± 0.02	4.69 ± 0.00	10.45 ± 0.82	814.72 ± 16.68	0.09 ± 0.02	42.53 ± 55.13	16.73
B131-G189	10.71	41.29	16.11 ± 0.01	15.07 ± 0.01	15.37 ± 0.02	5.45 ± 0.00	8.83 ± 1.01	821.85 ± 15.59	0.03 ± 0.01	37.90 ± 6.34	19.90
B132	10.71	41.26	17.36 ± 0.02	16.32 ± 0.02	16.57 ± 0.02	5.18 ± 0.00	12.19 ± 1.32	822.85 ± 8.89	0.08 ± 0.02	166.57 ± 57.94	19.38
B133	10.71	41.39	19.56 ± 0.00	18.78 ± 0.00	19.27 ± 0.00	4.45 ± 0.00	9.01 ± 0.67	815.29 ± 11.43	0.07 ± 0.01	0.75 ± 89.15	17.97
B134-G190	10.72	41.23	17.90 ± 0.02	16.86 ± 0.02	17.12 ± 0.01	4.90 ± 0.00	11.07 ± 0.90	828.48 ± 10.07	0.03 ± 0.02	160.92 ± 10.22	25.17
B136-G194	10.72	41.33	18.29 ± 0.02	17.27 ± 0.02	17.54 ± 0.02	4.94 ± 0.00	11.77 ± 0.72	817.77 ± 11.63	0.02 ± 0.01	10.17 ± 56.64	24.36
B138	10.73	41.31	18.75 ± 0.02	17.70 ± 0.02	17.97 ± 0.02	4.80 ± 0.00	11.54 ± 1.29	813.50 ± 13.64	0.07 ± 0.01	167.58 ± 63.68	23.20
B143-G198	10.75	41.32	18.20 ± 0.01	17.09 ± 0.01	17.29 ± 0.01	4.88 ± 0.00	8.60 ± 0.80	805.39 ± 12.80	0.06 ± 0.01	179.89 ± 59.60	24.76
B144	10.75	41.27	19.18 ± 0.03	18.05 ± 0.03	18.26 ± 0.03	4.77 ± 0.00	12.01 ± 1.33	803.55 ± 15.01	0.06 ± 0.01	6.63 ± 46.08	23.97
B145	10.76	41.21	20.11 ± 0.01	19.10 ± 0.02	19.38 ± 0.02	4.47 ± 0.00	12.77 ± 0.74	808.59 ± 12.59	0.06 ± 0.01	26.44 ± 80.69	19.93
B146	10.76	41.26	18.79 ± 0.01	17.75 ± 0.01	18.07 ± 0.01	4.69 ± 0.00	11.70 ± 0.60	824.70 ± 12.56	0.04 ± 0.01	27.46 ± 42.68	25.10
B147-G199	10.76	41.36	18.27 ± 0.01	17.07 ± 0.01	17.24 ± 0.02	4.89 ± 0.00	6.95 ± 0.95	795.71 ± 18.03	0.10 ± 0.01	72.82 ± 53.12	22.30
B148-G200	10.77	41.30	17.66 ± 0.02	16.61 ± 0.03	16.86 ± 0.03	4.94 ± 0.00	10.95 ± 1.08	816.05 ± 11.89	0.01 ± 0.03	156.61 ± 5.87	25.56
B150-G203	10.78	41.34	18.77 ± 0.03	17.69 ± 0.03	17.91 ± 0.04	4.71 ± 0.00	8.67 ± 1.38	832.75 ± 20.52	0.07 ± 0.02	21.66 ± 55.48	23.99
B151-G205	10.79	41.36	17.45 ± 0.02	16.09 ± 0.02	16.08 ± 0.02	5.25 ± 0.00	2.75 ± 0.73	811.63 ± 10.25	0.15 ± 0.01	109.72 ± 46.44	20.13
B152-G207	10.79	41.30	17.92 ± 0.01	16.80 ± 0.01	17.00 ± 0.02	4.98 ± 0.00	12.74 ± 0.87	810.39 ± 12.81	0.01 ± 0.03	118.19 ± 7.82	23.04
B153	10.79	41.25	18.72 ± 0.02	17.51 ± 0.02	17.63 ± 0.02	4.73 ± 0.00	10.07 ± 1.00	826.86 ± 13.00	0.05 ± 0.02	22.24 ± 10.31	20.69
...	...	...	...	...	...	...	...	...	...	...	...
SK086B	10.43	40.89	N/A	19.90 ± 0.04	20.68 ± 0.04	3.77 ± 0.01	13.94 ± 0.94	778.57 ± 11.64	0.06 ± 0.01	146.58 ± 6.12	14.63
SK107B	10.64	41.18	N/A	19.53 ± 0.01	19.79 ± 0.01	4.39 ± 0.00	11.44 ± 0.93	844.42 ± 12.87	0.08 ± 0.02	163.26 ± 2.71	21.59
V014	10.31	40.57	N/A	17.95 ± 0.02	18.49 ± 0.02	4.61 ± 0.00	7.07 ± 0.85	823.28 ± 13.84	0.09 ± 0.02	174.57 ± 14.02	9.41
V202	10.20	40.93	N/A	19.27 ± 0.02	19.68 ± 0.02	4.27 ± 0.00	12.55 ± 0.63	820.29 ± 11.02	0.08 ± 0.02	7.49 ± 51.13	15.66
WH2	10.00	40.56	N/A	21.42 ± 0.01	21.77 ± 0.01	3.63 ± 0.01	12.51 ± 0.76	804.16 ± 14.18	0.09 ± 0.01	173.43 ± 6.24	10.04

Notes. Names, coordinates, and π -DOC's estimates for various quantities: total integrated magnitudes in the F336W, F475W, and F814W passbands that are derived from the decontaminated images, total mass derived from mass map, age, distance, ellipticity, position angle, and the Mahalanobis distance to the training manifold. Where applicable, columns also indicate π -DOC's epistemic uncertainties (for physical uncertainties we refer to our 2σ confidence limits). The epistemic uncertainties are provided as separate columns in the online version of the table. GCs in the PHAST survey have no F336W magnitude counterpart and their estimates are inferred from π -DOC₂.